Module Manual

Master of Science (M.Sc.)

Data Science

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Program description

Content

The international master's program in data science is interdisciplinary and teaches a broad spectrum of methods for the representation, processing, modeling, provision, and storage of data. The program builds on the two pillars of mathematics and computer science and offers a variety of data science application fields in the third pillar. The program builds on the bachelor's degree in data science and is also open to bachelor's degree graduates in computer science or mathematics with relevant majors in data science, machine learning, or artificial intelligence.

The program is organized as a two-year program (four semesters) and begins each year in October. It consists of two and a half semesters of lectures and lab courses and one and a half semesters devoted to working in a research team (project work) and writing the master's thesis. The curriculum offers a lot of freedom to set your own focus in the field of data science. The academic degree of Master of Science is awarded. The language of study is English.

Graduates of the program are taught the fundamentals and knowledge required for successful work in the field of data science in an international environment. They acquire comprehensive knowledge of the mathematical and computer science foundations of this discipline and learn to practically apply the theoretical concepts in various application areas. Upon completion of the program, students are able to independently solve problems in the field of data science and related disciplines. Graduates are able to apply data science methods, critically examine results and further develop existing methods on the basis of new findings.

Career prospects

The master's program in data science optimally prepares graduates for a career in research and development in an academic or industrial environment. A data scientist typically works in an environment where large amounts of data are generated and is responsible for their analysis, algorithmic processing and feature extraction. He or she acquires knowledge in an application area and may work in an interdisciplinary team with application experts. A data scientist works in a research-oriented manner and is thus always up to date with the latest developments in this rapidly evolving field. Due to the high amount of computer science in the degree program, graduates are familiar with all the rules of software design, so that the career opportunities of classical computer scientists are also open to them. The degree program qualifies students for doctoral studies at a university.

Learning target

The master's program in data science is designed to prepare students for a professional career in research and development. The skills and knowledge required for this are taught as part of the degree program. In distinction to the bachelor's degree program in data science, the competencies mentioned here refer to complex problems, the consideration of uncertainty and working under given boundary conditions from application fields. The learning objectives of the program are achieved through the interaction of basic and advanced modules from data science, mathematics and computer science. The learning objectives are divided into the categories knowledge, skills, social competence and independence in the following.

Knowledge

Knowledge in data science is composed of theories and methods. It is acquired in the data science master's program in the following areas:

1. Graduates can explain the underlying mathematical, statistical, computer science and software engineering principles for the field of data science in depth and breadth.
2. Graduates have in-depth knowledge of statistical models and can compare different methods.
3. Graduates can describe methods for the representation, processing, modeling, provision and storage of large amounts of data and explain the principles of data management in depth.
4. Graduates have in-depth knowledge in the field of computer science and can describe principles of modern software development and place them in the context of current research results.
5. Graduates have in-depth knowledge in the field of machine learning and can explain data preparation, effective training and evaluation of trained models.
6. Graduates can classify their knowledge of different application areas of data science and combine it to solve complex problems as well as adapt and further develop the methodological knowledge they have learned to specific applications.
7. Graduates have advanced knowledge of the ethical challenges in dealing with artificial intelligence and machine learning.

Skills

1. The ability to apply the acquired knowledge to solve specific problems is fostered in the master's program data science in the following ways:
2. Graduates are able to systematically acquire data and store it in scalable data management systems.
3. Graduates are able to recognize patterns in unstructured data and to represent correlations appropriately.
4. Graduates are able to train data-driven models and can prepare data appropriately, select the model architecture appropriately, perform the training effectively, and evaluate the accuracy of the models appropriately.
5. Graduates are able to design and implement complex software systems.
6. Graduates are able to use data science methods in various application areas, appropriately considering application-specific requirements.

Social skills

1. Graduates are able to communicate with experts and laymen about scientific contents and problems in the field of data science. They can respond appropriately to questions, additions and comments and integrate feedback.
2. Graduates are able to take responsibility for the development of group results, to divide and distribute tasks for this purpose, to jointly coordinate procedures, to combine results and, if necessary, to present them jointly. They are able to develop suitable solution strategies in the event of difficulties in the group.

Autonomy

1. Graduates are able to present the scientific approach and the resulting results of their work in writing and orally in a comprehensible and technically appropriate manner.
2. Graduates are able to research necessary information, to put it into the context of their knowledge and to evaluate its relevance.
3. Graduates are able to realistically assess their existing competencies, independently compensate for deficits and independently acquire additional competencies.
4. Graduates are able to develop research areas in a self-organized and self-motivated manner and to find and define new problems (lifelong research).

Program structure

The curriculum of the master's program in data science is structured as follows:

Core Qualification (66 CP in total)

The core qualification contains the compulsory modules (each 6 CP):

- Advanced Machine Learning
- Linear Models and Regression
- Big Data
- Information Geometry
- Advanced Ethics of Machine Learning

In addition, the core qualification contains a seminar (3 CP), a lecture on scientific methods (3 CP), a research project (12 CP) as well as the interdisciplinary compulsory modules on operations & management (6 CP) and on non-technical courses (6 CP). As an additional module, the core qualification also contains a technical complementary course, where students can choose any module (6 CP) from the master courses of the TUHH.

Specializations (total 24 CP)

The curriculum contains the following three mandatory specializations:

- Specialization I. Data Science & Mathematics (12 CP in total)
- Specialization II. Computer Science (6 CP in total)
- Specialization III. Applocations (6 CP in total)

Master's thesis (30 CP, 4th semester)

The program is completed with the Master's thesis, which has a scope of 30 CP and is written in the 4th semester.

This results in a total workload of 120 LP.

Core Qualification

Module M0523: Business & Management

Module Responsible	Prof. Matthias Meyer
Admission Requirements	Successful completion of the modul "Foundations of Management"
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Students are able to find their way around selected special areas of management within the scope of business management. • Students are able to explain basic theories, categories, and models in selected special areas of business management. • Students are able to interrelate technical and management knowledge.
Skills	• Students are able to apply basic methods in selected areas of business management. • Students are able to explain and give reasons for decision proposals on practical issues in areas of business management.
Personal Competence	
Social Competence	• Students are able to communicate in small interdisciplinary groups and to jointly develop solutions for complex problems
Autonomy	• Students are capable of acquiring necessary knowledge independently by means of research and preparation of material.
Workload in Hours	Depends on choice of courses
Credit points	6

Courses
Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M1552: Advanced Machine Learning

Courses

Title	Typ	Hrs/wk	CP
Advanced Machine Learning (L2322)	Lecture	2	3
Advanced Machine Learning (L2323)	Recitation Section (small)	2	3

Module Responsible	Dr. Jens-Peter Zemke
Admission Requirements	None
Recommended Previous Knowledge	1. Mathematics I-III 2. Numerical Mathematics 1/ Numerics 3. Programming skills, preferably in Python
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students are able to name, state and classify state-of-the-art neural networks and their corresponding mathematical basics. They can assess the difficulties of different neural networks.
Skills	Students are able to implement, understand, and, tailored to the field of application, apply neural networks.
Personal Competence	
Social Competence	Students can • develop and document joint solutions in small teams; • form groups to further develop the ideas and transfer them to other areas of applicability; • form a team to develop, build, and advance a software library.
Autonomy	Students are able to • correctly assess the time and effort of self-defined work; • assess whether the supporting theoretical and practical excercises are better solved individually or in a team; • define test problems for testing and expanding the methods; • assess their individual progess and, if necessary, to ask questions and seek help.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computational Methods and Machine Learning in Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L2322: Advanced Machine Learning

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	1. Basics: analogy; layout of neural nets, universal approximation, NP-completeness 2. Feedforward Neural Networks: backpropagation, variants of Stochastic Gradients 3. Convolutional Neural Networks: idea, layout, FFT and Winograds algorithms, implementation details 4. Adversarial Attacks 5. Recurrent Neural Networks: idea, dynamical systems, training, LSTM 6. Residual Neural Networks 7. Neural ODEs 8. Autoencoder and Generative Adversarial Networks 9. Attention and Transformers
Literature	1. Skript 2. Online-Werke: ◦ http://neuralnetworksanddeeplearning.com/ ◦ https://www.deeplearningbook.org/

Course L2323: Advanced Machine Learning	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0524: Non-technical Courses for Master

Module Responsible	Dagmar Richter
Admission Requirements	None
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	**The Nontechnical Academic Programms (NTA)** imparts skills that, in view of the TUHH's training profile, professional engineering studies require but are not able to cover fully. Self-reliance, self-management, collaboration and professional and personnel management competences. The department implements these training objectives in its **teaching architecture**, in its **teaching and learning arrangements**, in **teaching areas** and by means of teaching offerings in which students can qualify by opting for **specific competences** and a **competence level** at the Bachelor's or Master's level. The teaching offerings are pooled in two different catalogues for nontechnical complementary courses. **The Learning Architecture** consists of a cross-disciplinarily study offering. The centrally designed teaching offering ensures that courses in the nontechnical academic programms follow the specific profiling of TUHH degree courses. The learning architecture demands and trains independent educational planning as regards the individual development of competences. It also provides orientation knowledge in the form of "profiles". The subjects that can be studied in parallel throughout the student's entire study program - if need be, it can be studied in one to two semesters. In view of the adaptation problems that individuals commonly face in their first semesters after making the transition from school to university and in order to encourage individually planned semesters abroad, there is no obligation to study these subjects in one or two specific semesters during the course of studies. **Teaching and Learning Arrangements** provide for students, separated into B.Sc. and M.Sc., to learn with and from each other across semesters. The challenge of dealing with interdisciplinarity and a variety of stages of learning in courses are part of the learning architecture and are deliberately encouraged in specific courses. **Fields of Teaching** are based on research findings from the academic disciplines cultural studies, social studies, arts, historical studies, communication studies, migration studies and sustainability research, and from engineering didactics. In addition, from the winter semester 2014/15 students on all Bachelor's courses will have the opportunity to learn about business management and start-ups in a goal-oriented way. The fields of teaching are augmented by soft skills offers and a foreign language offer. Here, the focus is on encouraging goal-oriented communication skills, e.g. the skills required by outgoing engineers in international and intercultural situations. **The Competence Level** of the courses offered in this area is different as regards the basic training objective in the Bachelor's and Master's fields. These differences are reflected in the practical examples used, in content topics that refer to different professional application contexts, and in the higher scientific and theoretical level of abstraction in the B.Sc. This is also reflected in the different quality of soft skills, which relate to the different team positions and different group leadership functions of Bachelor's and Master's graduates in their future working life. **Specialized Competence (Knowledge)** Students can • explain specialized areas in context of the relevant non-technical disciplines, • outline basic theories, categories, terminology, models, concepts or artistic techniques in the disciplines represented in the learning area, • different specialist disciplines relate to their own discipline and differentiate it as well as make connections, • sketch the basic outlines of how scientific disciplines, paradigms, models, instruments, methods and forms of representation in the specialized sciences are subject to individual and socio-cultural interpretation and historicity, • Can communicate in a foreign language in a manner appropriate to the subject.
Skills	**Professional Competence (Skills)** In selected sub-areas students can • apply basic and specific methods of the said scientific disciplines, • aquestion a specific technical phenomena, models, theories from the viewpoint of another, aforementioned specialist discipline, • to handle simple and advanced questions in aforementioned scientific disciplines in a sucsessful manner, • justify their decisions on forms of organization and application in practical questions in contexts that go beyond the technical relationship to the subject.
Personal Competence	
Social Competence	**Personal Competences (Social Skills)**

	Students will be able • to learn to collaborate in different manner, • to present and analyze problems in the abovementioned fields in a partner or group situation in a manner appropriate to the addressees, • to express themselves competently, in a culturally appropriate and gender-sensitive manner in the language of the country (as far as this study-focus would be chosen), • to explain nontechnical items to auditorium with technical background knowledge.
Autonomy	**Personal Competences (Self-reliance)** Students are able in selected areas • to reflect on their own profession and professionalism in the context of real-life fields of application • to organize themselves and their own learning processes • to reflect and decide questions in front of a broad education background • to communicate a nontechnical item in a competent way in writen form or verbaly • to organize themselves as an entrepreneurial subject country (as far as this study-focus would be chosen)
Workload in Hours	Depends on choice of courses
Credit points	6

Courses
Information regarding lectures and courses can be found in the corresponding module handbook published separately.

Module M2071: Linear Models and Regression

Courses

Title	Typ	Hrs/wk	CP
Linear Models and Regression (L3343)	Lecture	3	4
Linear Models and Regression (L3344)	Recitation Section (small)	1	2

Module Responsible	Prof. Matthias Schulte
Admission Requirements	None
Recommended Previous Knowledge	Preknowledge in probability and statistics
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Students know the fundamental models for regression and corresponding tests and are able to explain them using appropriate examples. • Students can discuss logical connections between these concepts and are capable of illustrating these connections with the help of examples. • Students know proof strategies and can reproduce them.
Skills	• Students can investigate statistical problems with the help of the models studied in the course. • Students are able to explore and verify further logical connections between the concepts studied in the course. • For a given problem, the students can develop and execute a suitable approach, and are able to critically evaluate the results.
Personal Competence	
Social Competence	• Students are able to work together (e.g. on their regular home work) and to present their results appropriately (e.g. during exercise class). • In doing so, they can communicate new concepts and they can design examples to check and deepen the understanding of their peers.
Autonomy	• Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. • Students can put their knowledge in relation to the contents of other lectures. • Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L3343: Linear Models and Regression	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	WiSe
Content	- Linear models - Testing in the Gaussian linear model - Analysis of variance - Asymptotic behaviour of maximum likelihood estimators - Nonlinear regression - Logistic and Poisson regression - Generalised linear models - Linear discriminant analysis - Principal component analysis
Literature	P. Dunn and G. Smyth (2018): Generalized linear models with examples in R. Springer. L. Fahrmeir, T. Kneib, S. Lang and B. Marx (2021): Regression - models, methods and applications. Second edition, Springer.

Course L3344: Linear Models and Regression	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1871: Big Data

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Big Data (L3101)	Lecture	2	3
Big Data (L3102)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Stefan Schulte
Admission Requirements	None
Recommended Previous Knowledge	• Good software engineering and implementation skills are very important for the practical part of this module • Previous knowledge in the fields of Distributed Systems and Databases is helpful
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	After successful completion of the course, students are able to: • Describe basic concepts of methods and technologies to process and store very large amounts of data • Discuss and assess critical aspects of Big Data • Select and apply Big Data technologies for particular application areas • Design and develop practical solutions for the processing of very large amounts of data • Implement Big Data services
Skills	The students acquire the ability to model Big Data systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies.
Personal Competence	
Social Competence	Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem.
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Group project incl. presentation, written exam
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory

Course L3101: Big Data

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses the fundamental concepts of modern Big Data systems. The following topics will be covered in the single lectures: • Cloud Computing • Data models for Big Data • NoSQL databases • Lambda architecture including batch, speed, and serving layers • Kappa architecture • Concepts and tools for data stream processing • Current software tools for the processing of very large amounts of data
Literature	Lecture notes as well as current literature, are announced in the lecture.

Course L3102: Big Data	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture with a concrete technical problem, which needs to be solved by the students in group work during the semester. The topics are from the field of Big Data, and address for instance data stream processing or batch processing.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M2072: Information Geometry

Courses

Title	Typ	Hrs/wk	CP
Information Geometry (L3383)	Lecture	2	3
Information Geometry (L3384)	Recitation Section (small)	3	3

Module Responsible	Prof. Nihat Ay
Admission Requirements	None
Recommended Previous Knowledge	Linear Algebra, Analysis, Fundamentals of Probability Theory, Basic Programming Course
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students know: - statistical models - geometric concepts for statistical models - spherical geometry - Fisher-Rao metric and Fisher information - Shannon entropy - maximum entropy method - exponential family - Kullback-Leibler divergence and information projections - statistical estimators and Cramér-Rao inequality - Fisher efficiency of statistical estimators - Maximum-Likelihood method - the natural gradient for learning algorithms - graphical models and probabilistic neural networks - Gibbs-sampling and generative models
Skills	The students are able to: - apply geometric concepts in the context of probability theory - evaluate geometric quantities in the context of probability theory - apply the maximum entropy method - apply efficient statistical estimation - derive and apply learning algorithms based on the natural gradient - train generative neural networks
Personal Competence	
Social Competence	Students are able to work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve problems.
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Core Qualification: Compulsory

Course L3383: Information Geometry	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Nihat Ay
Language	EN
Cycle	SoSe
Content	In this lecture, unifying and natural geometric concepts for probability theory are developed and formalized. The approach is designed in a suggestive manner and illustrated using elementary examples. The application of the resulting formalism is demonstrated using examples in the context of statistics, the theory of graphical models, and machine learning. Specific contents: - statistical models - geometric concepts for statistical models - spherical geometry - Fisher-Rao metric and Fisher information - Shannon entropy - maximum entropy method - exponential family - Kullback-Leibler divergence and information projections - statistical estimators and Cramér-Rao inequality - Fisher efficiency of statistical estimators - Maximum-Likelihood method - the natural gradient for learning algorithms - graphical models and probabilistic neural networks - Gibbs-sampling and generative models
Literature	- S. Amari, H. Nagaoka: Methods of Information Geometry. Oxford University Press, 2000. - S. Amari: Information Geometry and Its Applications. Springer, 2016. - N. Ay, J. Jost, L. Schwachhöfer, H. V. Lê. Information Geometry, 2017. - S. Lauritzen: Graphical Models. Oxford University Press, 1996 (reprinted 2004). - J. Pearl: Causality: Models, Reasoning and Inference. Second edition, Cambridge University Press, 2009.

Course L3384: Information Geometry	
Typ	Recitation Section (small)
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Nihat Ay, Adwait Datar
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1563: Research Project Computer Science

Courses

Title	Typ	Hrs/wk	CP
Research Project Computer Science (L2353)	Projection Course	8	12

Module Responsible	Dozenten des SD E
Admission Requirements	None
Recommended Previous Knowledge	Basic knowledge and techniques from the Master courses in the semesters 1 and 2.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students are able to acquire advanced knowledge in a subfield of Computer Science and can independently acquire deeper knowledge in the field.
Skills	The students are able to formulate the scientific problems to be considered and to work out solutions in an independent manner and to realize them.
Personal Competence	
Social Competence	The students are able to discuss proposals for solutions of scientific problems within the team. They are able to present the results in a clear and well structured manner.
Autonomy	The students can provide a scientific work in a timely manner and document the results in a detailed and well readable form. They are able to actively follow anticipate the presentations of other students such that eventually a scientific discussion comes up.
Workload in Hours	Independent Study Time 248, Study Time in Lecture 112
Credit points	12
Course achievement	None
Examination	Study work
Examination duration and scale	Vortrag
Assignment for the Following Curricula	Computer Science: Core Qualification: Compulsory Data Science: Core Qualification: Compulsory

Course L2353: Research Project Computer Science

Typ	Projection Course
Hrs/wk	8
CP	12
Workload in Hours	Independent Study Time 248, Study Time in Lecture 112
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe
Content	Current research topics of the chosen areas of specialization
Literature	Wird vom Veranstalter bekanntgegeben.

Module M1873: Scientific Methods

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Scientific Methods (L3088)	Lecture	2	3

Module Responsible	Prof. Stefan Schulte
Admission Requirements	None
Recommended Previous Knowledge	Students should have first experiences with the principles of research work. Usually, this was acquired while working on the Bachelor thesis or in seminars during the Bachelor studies. We strongly recommend to first do the module "Research Methods", and then the Master seminar.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Systematic literature reviews • Experimental software engineering • Design science • Good scientific practice • Scientific writing • Presenting scientific results using LaTeX • Visualization of scientific results
Skills	After successful completion of the course, students are able to: • Assess which research methods from the fields of Computer Science, Data Science and related disciplines are suitable for which problems, • describe and apply the steps necessary to carry out a particular research method, • structure a scientific text, • review and cite scientific sources, • apply Best Practice scientific behavior, • conduct evaluations and visualize their results, and • apply scientific methods within their Master theses.
Personal Competence	
Social Competence	Students are able to work on complex scientific questions and to present their results.
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Credit points	3
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Group project (written elaboration)
Assignment for the Following Curricula	Data Science: Core Qualification: Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory

Course L3088: Scientific Methods	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe
Content	This university course on Scientific Methods in computer science, data science, and related fields, explores methods for systematic literature reviews, experimental software engineering, and design science. It emphasizes good scientific practice, skills in scientific writing, and presenting results. Students will learn effective visualization techniques for scientific outcomes, preparing them for rigorous academic and research environments.
Literature	Lecture notes as well as current literature, are announced in the lecture.

Module M1921: Technical Complementary Course for DSMS (according to Subject Specific Regulations)

Courses			
Title	**Typ**	**Hrs/wk**	**CP**

Module Responsible	Prof. Stefan Schulte
Admission Requirements	None
Recommended Previous Knowledge	See selected module according to Subject Specific Regulations
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	See selected module according to Subject Specific Regulations
Skills	See selected module according to Subject Specific Regulations
Personal Competence	
Social Competence	See selected module according to Subject Specific Regulations
Autonomy	See selected module according to Subject Specific Regulations
Workload in Hours	Depends on choice of courses
Credit points	6
Assignment for the Following Curricula	Data Science: Core Qualification: Compulsory

Module M2073: Advanced Ethics of Machine Learning

Courses

Title	Typ	Hrs/wk	CP
Advanced Ethics of Machine Learning (L3399)	Lecture	2	3
Case Studies in Advanced Ethics of Machine Learning (L3400)	Seminar	2	3

Module Responsible	Prof. Maximilian Kiener
Admission Requirements	None
Recommended Previous Knowledge	Basics of machine learning
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Familiarity with some ethical questions in the use of AI
Skills	
Personal Competence	
Social Competence	--
Autonomy	--
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	x
Assignment for the Following Curricula	Data Science: Core Qualification: Compulsory

Course L3399: Advanced Ethics of Machine Learning

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Maximilian Kiener
Language	EN
Cycle	WiSe
Content	
Literature	

Course L3400: Case Studies in Advanced Ethics of Machine Learning

Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Maximilian Kiener
Language	EN
Cycle	WiSe
Content	
Literature	

Module M2115: Advanced Seminar EIM I

Courses

Title	Typ	Hrs/wk	CP
Advanced Seminar EIM I (L3385)	Seminar	2	3

Module Responsible	Dozenten des SD E
Admission Requirements	None
Recommended Previous Knowledge	Basic knowledge in the area of the seminar within the fields of computer science, communication technology, electrical engineering or mathematics at the Master's level.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students are able to • explicate a specific topic from the field of the seminar, • describe complex issues, • present different views and evaluate in a critical way.
Skills	The students are able to • familiarize in a specific topic in limited time, • realize a literature survey on the specific topic and cite in a correct way, • elaborate a presentation and give a lecture to a selected audience, • sum up the presentation in 10-15 lines, • answer questions in the final discussion.
Personal Competence	
Social Competence	The students are able to • elaborate and introduce a topic for a certain audience, • discuss the topic, content and structure of the presentation with the instructor, • discuss certain aspects with the audience, and • as the lecturer listen and respond to questions from the audience.
Autonomy	The students are able to • define the task in question in an autonomous way, • develop the necessary knowledge, • use appropriate work equipment, and • guided by an instructor critically check the working status.
Workload in Hours	Depends on choice of courses
Credit points	3
Assignment for the Following Curricula	Computer Science: Specialisation IV. Subject Specific Focus: Elective Compulsory Data Science: Core Qualification: Compulsory Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Compulsory Electrical Engineering and Information Technology: Specialisation Nanoelectronics and Microsystems Technology: Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Compulsory Computer Science in Engineering: Specialisation IV. Subject Specific Focus: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory

Course L3385: Advanced Seminar EIM I

Typ	Seminar
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	EN
Cycle	WiSe/SoSe
Content	- Seminar presentations by enrolled students about selected topics of computer science, communication technology, electrical engineering or mathematics - Active participation in discussions
Literature	Wird vom Veranstalter bekanntgegeben.

Specialization I. Data Science & Mathematics

Module M0716: Hierarchical Algorithms

Courses

Title	Typ	Hrs/wk	CP
Hierarchical Algorithms (L0585)	Lecture	2	3
Hierarchical Algorithms (L0586)	Recitation Section (small)	2	3

Module Responsible	Prof. Sabine Le Borne
Admission Requirements	None
Recommended Previous Knowledge	• Mathematics I, II, III for Engineering students (german or english) or Analysis & Linear Algebra I + II as well as Analysis III for Technomathematicians • Numerical Mathematics 1 • Programming experience in C
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students are able to • name representatives of hierarchical algorithms and list their characteristics, • explain construction techniques for hierarchical algorithms, • discuss aspects regarding the efficient implementation of hierarchical algorithms.
Skills	Students are able to • implement the hierarchical algorithms discussed in the lecture, • analyse the storage and computational complexities of the algorithms, • adapt algorithms to problem settings of various applications and thus develop problem adapted variants.
Personal Competence	
Social Competence	Students are able to • work together in heterogeneously composed teams (i.e., teams from different study programs and background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms.
Autonomy	Students are capable • to assess whether the supporting theoretical and practical excercises are better solved individually or in a team, • to work on complex problems over an extended period of time, • to assess their individual progess and, if necessary, to ask questions and seek help.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Oral exam
Examination duration and scale	20 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory

Course L0585: Hierarchical Algorithms

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sabine Le Borne
Language	EN
Cycle	WiSe
Content	• Low rank matrices • Separable expansions • Hierarchical matrix partitions • Hierarchical matrices • Formatted matrix operations • H2 matrices • Applications • Additional topics (e.g. matrix functions, tensor products)
Literature	W. Hackbusch: Hierarchical matrices: Algorithms and Analysis, Springer (2015)

Course L0586: Hierarchical Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sabine Le Borne
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1428: Linear and Nonlinear Optimization

Courses

Title	Typ	Hrs/wk	CP
Linear and Nonlinear Optimization (L2062)	Lecture	4	4
Linear and Nonlinear Optimization (L2063)	Recitation Section (large)	1	2

Module Responsible	Prof. Matthias Mnich
Admission Requirements	None
Recommended Previous Knowledge	• Discrete Algebraic Structures • Mathematics I • Graph Theory and Optimization
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Students can name the basic concepts in linear and non-linear optimization. They are able to explain them using appropriate examples. • Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. • They know proof strategies and can reproduce them.
Skills	• Students can model problems in linear and non-linear optimization with the help of the concepts studied in this course. Moreover, they are capable of solving them by applying established methods. • Students are able to discover and verify further logical connections between the concepts studied in the course. • For a given problem, the students can develop and execute a suitable approach, and are able to critically evaluate the results.
Personal Competence	
Social Competence	• Students are able to work together in teams. They are capable to use mathematics as a common language. • In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers.
Autonomy	• Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. • Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** No; **Bonus** 20 %; **Form** Excercises; **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Energy Systems: Elective Compulsory

Course L2062: Linear and Nonlinear Optimization	
Typ	Lecture
Hrs/wk	4
CP	4
Workload in Hours	Independent Study Time 64, Study Time in Lecture 56
Lecturer	Prof. Matthias Mnich
Language	DE/EN
Cycle	WiSe
Content	• Modelling linear programming problems • Graphical method • Algebraic background • Convexity • Polyhedral theory • Simplex method • Degeneracy and convergence • duality • interior-point methods • quadratic optimization • integer linear programming
Literature	• A. Schrijver: Combinatorial Optimization: Polyhedra and Efficiency. Springer, 2003 • B. Korte and T. Vygen: Combinatorial Optimization: Theory and Algorithms. Springer, 2018 • T. Cormen, Ch. Leiserson, R. Rivest, C. Stein: Introduction to Algorithms. MIT Press, 2013

Course L2063: Linear and Nonlinear Optimization	
Typ	Recitation Section (large)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Matthias Mnich
Language	DE/EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0720: Matrix Algorithms

Courses

Title	Typ	Hrs/wk	CP
Matrix Algorithms (L0984)	Lecture	2	3
Matrix Algorithms (L0985)	Recitation Section (small)	2	3

Module Responsible	Dr. Jens-Peter Zemke
Admission Requirements	None
Recommended Previous Knowledge	• Mathematics I - III • Numerical Mathematics 1/ Numerics • Basic knowledge of the programming language Python
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students are able to 1. name, state and classify state-of-the-art Krylov subspace methods for the solution of the core problems of the engineering sciences, namely, eigenvalue problems, solution of linear systems, and model reduction; 2. state approaches for the solution of matrix equations (Sylvester, Lyapunov, Riccati).
Skills	Students are capable to 1. implement and assess basic Krylov subspace methods for the solution of eigenvalue problems, linear systems, and model reduction; 2. assess methods used in modern software with respect to computing time, stability, and domain of applicability; 3. adapt the approaches learned to new, unknown types of problem.
Personal Competence	
Social Competence	Students can • develop and document joint solutions in small teams; • form groups to further develop the ideas and transfer them to other areas of applicability; • form a team to develop, build, and advance a software library.
Autonomy	Students are able to • correctly assess the time and effort of self-defined work; • assess whether the supporting theoretical and practical excercises are better solved individually or in a team; • define test problems for testing and expanding the methods; • assess their individual progess and, if necessary, to ask questions and seek help.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Oral exam
Examination duration and scale	25 min
Assignment for the Following Curricula	Computational Methods and Machine Learning in Engineering: Core Qualification: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Simulation Technology: Elective Compulsory

Course L0984: Matrix Algorithms	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	• Part A: Krylov Subspace Methods: ∘ Basics (derivation, basis, Ritz, OR, MR) ∘ Arnoldi-based methods (Arnoldi, GMRes) ∘ Lanczos-based methods (Lanczos, CG, BiCG, QMR, SymmLQ, PvL) ∘ Sonneveld-based methods (IDR, BiCGStab, TFQMR, IDR(s)) • Part B: Matrix Equations: ∘ Sylvester Equation ∘ Lyapunov Equation ∘ Algebraic Riccati Equation
Literature	Skript (224 Seiten) Ergänzend können die folgenden Lehrbücher herangezogen werden: 1. Saad, Yousef. Numerical methods for large eigenvalue problems: revised edition. Society for Industrial and Applied Mathematics, 2011. 2. Saad, Yousef. Iterative methods for sparse linear systems. Society for Industrial and Applied Mathematics, 2003. 3. Van der Vorst, Henk A. Iterative Krylov methods for large linear systems. No. 13. Cambridge University Press, 2003. 4. Liesen, Jörg, and Zdenek Strakos. Krylov subspace methods: principles and analysis. Oxford University Press, 2013.

Course L0985: Matrix Algorithms	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0714: Numerical Methods for Ordinary Differential Equations

Courses

Title	Typ	Hrs/wk	CP
Numerical Treatment of Ordinary Differential Equations (L0576)	Lecture	2	3
Numerical Treatment of Ordinary Differential Equations (L0582)	Recitation Section (small)	2	3

Module Responsible	Prof. Daniel Ruprecht
Admission Requirements	None
Recommended Previous Knowledge	• Mathematik I, II, III for Engineers (German or English) or Analysis & Linear Algebra I + II plus Analysis III for Technomathematiker. • Basic knowledge of MATLAB, Python or a similar programming language.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students are able to • name numerical methods for the solution of ordinary differential equations and explain their core ideas, • formulate convergence statements for the taught numerical methods (including the necessary assumptions about the solved problem), • explain aspects regarding the practical realisation of a method, • select the appropriate numerical method for specific problems, implement the numerical algorithms efficiently and interpret the numerical results.
Skills	Students are able to • implement, apply and compare numerical methods for the solution of ordinary differential equations, • explain the convergence behaviour of numerical methods, taking into consideration the solved problem and selected algorithm, • develop a suitable solution approach for a given problem, if necessary by combining multiple algorithms, realise this approach and critically evaluate results.
Personal Competence	
Social Competence	Students are able to • work together in heterogeneous teams (i.e., teams from different study programs and with different background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms.
Autonomy	Students are capable • to assess whether the provided theoretical and practical excercises are better solved individually or in a team and • to assess their individual progress and, if necessary, to ask questions and seek help.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Bioprocess Engineering: Specialisation A - General Bioprocess Engineering: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Chemical and Bioprocess Engineering: Technical Complementary Course: Elective Compulsory Computational Methods and Machine Learning in Engineering: Core Qualification: Elective Compulsory Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Aircraft Systems Engineering: Core Qualification: Elective Compulsory Interdisciplinary Mathematics: Specialisation II. Numerical - Modelling Training: Compulsory Mechatronics: Core Qualification: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Compulsory Process Engineering: Specialisation Chemical Process Engineering: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory

Course L0576: Numerical Treatment of Ordinary Differential Equations	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	SoSe
Content	Numerical methods for Initial Value Problems • single step methods • multistep methods • stiff problems • differential algebraic equations (DAE) of index 1 Numerical methods for Boundary Value Problems • multiple shooting method • difference methods
Literature	• E. Hairer, S. Noersett, G. Wanner: Solving Ordinary Differential Equations I: Nonstiff Problems. • E. Hairer, G. Wanner: Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems. • D. Griffiths, D. Higham: Numerical Methods for Ordinary Differential Equations.

Course L0582: Numerical Treatment of Ordinary Differential Equations	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Daniel Ruprecht
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0711: Numerical Mathematics II

Courses

Title	Typ	Hrs/wk	CP
Numerical Mathematics II (L0568)	Lecture	2	3
Numerical Mathematics II (L0569)	Recitation Section (small)	2	3

Module Responsible	Dr. Jens-Peter Zemke
Admission Requirements	None
Recommended Previous Knowledge	• Numerical Mathematics I • Python knowledge
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students are able to • name advanced numerical methods for interpolation, approximation, integration, eigenvalue problems, eigenvalue problems, nonlinear root finding problems and explain their core ideas, • repeat convergence statements for the numerical methods, sketch convergence proofs, • explain practical aspects of numerical methods concerning runtime and storage needs • explain aspects regarding the practical implementation of numerical methods with respect to computational and storage complexity.
Skills	Students are able to • implement, apply and compare advanced numerical methods in Python, • justify the convergence behaviour of numerical methods with respect to the problem and solution algorithm and to transfer it to related problems, • for a given problem, develop a suitable solution approach, if necessary through composition of several algorithms, to execute this approach and to critically evaluate the results
Personal Competence	
Social Competence	Students are able to • work together in heterogeneously composed teams (i.e., teams from different study programs and background knowledge), explain theoretical foundations and support each other with practical aspects regarding the implementation of algorithms.
Autonomy	Students are capable • to assess whether the supporting theoretical and practical excercises are better solved individually or in a team, • to assess their individual progess and, if necessary, to ask questions and seek help.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Oral exam
Examination duration and scale	25 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory Theoretical Mechanical Engineering: Core Qualification: Elective Compulsory

Course L0568: Numerical Mathematics II	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	DE/EN
Cycle	SoSe
Content	1. Error and stability: Notions and estimates 2. Rational interpolation and approximation 3. Multidimensional interpolation (RBF) and approximation (neural nets) 4. Quadrature: Gaussian quadrature, orthogonal polynomials 5. Linear systems: Perturbation theory of decompositions, structured matrices 6. Eigenvalue problems: LR-, QD-, QR-Algorithmus 7. Nonlinear systems of equations: Newton and Quasi-Newton methods, line search (optional) 8. Krylov space methods: Arnoldi-, Lanczos methods (optional)
Literature	• Skript • Stoer/Bulirsch: Numerische Mathematik 1, Springer • Dahmen, Reusken: Numerik für Ingenieure und Naturwissenschaftler, Springer

Course L0569: Numerical Mathematics II	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Jens-Peter Zemke
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1405: Randomised Algorithms and Random Graphs

Courses

Title	Typ	Hrs/wk	CP
Randomised Algorithms and Random Graphs (L2010)	Lecture	2	3
Randomised Algorithms and Random Graphs (L2011)	Recitation Section (large)	2	3

Module Responsible	Prof. Anusch Taraz
Admission Requirements	None
Recommended Previous Knowledge	
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Students can describe basic concepts in the area of Randomized Algorithms and Random Graphs such as random walks, tail bounds, fingerprinting and algebraic techniques, first and second moment methods, and various random graph models. They are able to explain them using appropriate examples. • Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. • They know proof strategies and can apply them.
Skills	• Students can model problems with the help of the concepts studied in this course. Moreover, they are capable of solving them by applying established methods. • Students are able to explore and verify further logical connections between the concepts studied in the course. • For a given problem, the students can develop and execute a suitable technique, and are able to critically evaluate the results.
Personal Competence	
Social Competence	• Students are able to work together in teams. They are capable to establish a common language. • In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers.
Autonomy	• Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. • Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Oral exam
Examination duration and scale	30 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Computer Science in Engineering: Specialisation III. Mathematics: Elective Compulsory

Course L2010: Randomised Algorithms and Random Graphs	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Anusch Taraz, Prof. Volker Turau
Language	DE/EN
Cycle	SoSe
Content	Randomized Algorithms: • introduction and recalling basic tools from probability • randomized search • random walks • text search with fingerprinting • parallel and distributed algorithms • online algorithms Random Graphs: • typical properties • first and second moment method • tail bounds • thresholds and phase transitions • probabilistic method • models for complex networks
Literature	• Motwani, Raghavan: Randomized Algorithms • Worsch: Randomisierte Algorithmen • Dietzfelbinger: Randomisierte Algorithmen • Bollobas: Random Graphs • Alon, Spencer: The Probabilistic Method • Frieze, Karonski: Random Graphs • van der Hofstad: Random Graphs and Complex Networks

Course L2011: Randomised Algorithms and Random Graphs	
Typ	Recitation Section (large)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Anusch Taraz, Prof. Volker Turau
Language	DE/EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2060: Applied Time-Series Analysis and Forecasting

Courses

Title	Typ	Hrs/wk	CP
Applied time-series analysis and forecasting (Angewandte Zeitreihenanalyse und -prognose) (L3316)	Lecture	2	3
Applied time-series analysis and forecasting (Angewandte Zeitreihenanalyse und -prognose) (L3317)	Recitation Section (small)	2	3

Module Responsible	Dr. Prashant Batra
Admission Requirements	None
Recommended Previous Knowledge	Mathematics I-III (Real analysis,computing in vector spaces , principle of complete induction) ; Stochastics.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students know basic models for time series, such as AR (autoregressive), MA (moving average), ARMA and RW (random walk) processes, and their theory, and are familiar with the modeling of economic time series using the additive or multiplicative model. Students know and master simple descriptive and exploratory methods for time series: smoothing, differentiation, auto- and cross-correlation. Students know the basic terms such as (non)-stationarity, covariance, volatility, unbiased estimation and the analysis of time series in the frequency domain.
Skills	Students acquire skills in the analysis and modeling of empirical time series, can adapt basic models to real data using the statistical software R, and master model selection and application. Students can apply estimation and test theory, select an optimal model and determine confidence and prediction intervals for future observations. Students can develop a suitable solution approach to given problems, pursue it and critically evaluate the results. Students can use the acquired knowledge and skills for their own empirical work (e.g. bachelor's thesis).
Personal Competence	
Social Competence	Students form working groups independently and are able to discuss with other participants their suggestions for solutions of the tasks presented. Students learn to present their suggestions coherently to a group. Students are able to work together in teams and master mathematics as a common language.
Autonomy	
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Data Science: Specialisation I. Mathematics: Elective Compulsory

Course L3316: Applied time-series analysis and forecasting (Angewandte Zeitreihenanalyse und -prognose)	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Prashant Batra
Language	EN
Cycle	SoSe
Content	Examples of time series. Decomposition into/extraction of trend, seasonal, cyclic components; Time series regression and exploratory data analysis. Parametric methods: estimation/elimination of trend, seasonal and cyclic components. Estimation of models, model selection, ordering of models, and application. Weakly stationary processes, Gaussian white noise, i.i.d. noise, random walk processes. Non-parametric methods: autocovariance, (partial) autocorrelation. Unbiased estimation of theoretical moments. Hypothesis testing. Theory of autoregressive (AR), moving average (MA), random walk (RW), and ARMA(p,q) processes. Model selection for predictions using the least squares approach or maximum likelihood estimation. Yule-Walker equations, Durbin-Levinson algorithm, innovation algorithm. Exponential smoothing and transformation of time series. Unit root tests. Integrated models for non-stationary time series (ARIMA). Time series models for financial market data: Generalized AR models with fluctuating variance (GARCH). Time-continuous models e.g. geometric Brownian motion. Statistical methods in the frequency domain, state space models, Kalman filters..
Literature	R.H. Shumway/A. Stoffer: Time Series Analysis and its application 4th ed. , Springer, 2019 https://doi.org/10.1007/978-3-319-52452-8 Schlittgen, Rainer. Angewandte Zeitreihenanalyse mit R, Berlin, München, Boston: De Gruyter Oldenbourg, 2015 https://doi.org/10.1515/9783110413991 P.J. Brockwell/R.A. Davis: Introduction to Time Series Analysis and Forecasting, 3ed, Springer, 2016. A.V. Metcalfe; P. S.P. Cowpertwait : Introductory Time Series with R, Springer, 2009.https://doi.org/10.1007/978-0-387-88698-5 J. D. Cryer; K.-S. Chan: Time Series Analysis With Applications in R Second Edition, 2008 , Springer https://doi.org/10.1007/978-0-387-75959-3 J.-P. Kreiss; G.Neuhaus, Einführung in die Zeitreihenanalyse, Springer, 2006. https://doi.org/10.1007/3-540-33571-4 Ruppert, David: Statistics and Finance, Springer, 2004 https://doi.org/10.1007/978-1-4419-6876-0 Tsay, Ruey S.: Analysis of Financial Time Series, 2nd edition, Wiley, 2005 Carmona, René : Statistical Analysis of Financial Data in R, Second Edition, Springer,2014 https://link.springer.com/book/10.1007/978-1-4614-8788-3

Course L3317: Applied time-series analysis and forecasting (Angewandte Zeitreihenanalyse und -prognose)	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Prashant Batra
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1668: Probability Theory

Courses

Title	Typ	Hrs/wk	CP
Probability Theory (L2643)	Lecture	3	4
Probability Theory (L2644)	Recitation Section (small)	1	2

Module Responsible	Prof. Matthias Schulte
Admission Requirements	None
Recommended Previous Knowledge	Familiarity with the basic concepts of probability
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Students can name the basic concepts in probability theory. They are able to explain them using appropriate examples. • Students can discuss logical connections between these concepts. They are capable of illustrating these connections with the help of examples. • They know proof strategies and can reproduce them.
Skills	• Students can model problems from probability theory with the help of the concepts studied in this course. Moreover, they are capable of solving them by applying established methods. • Students are able to explore and verify further logical connections between the concepts studied in the course. • For a given problem, the students can develop and execute a suitable technique, and are able to critically evaluate the results.
Personal Competence	
Social Competence	• Students are able to work together (e.g. on their regular home work) and to present their results appropriately (e.g. during exercise class). • In doing so, they can communicate new concepts according to the needs of their cooperating partners. Moreover, they can design examples to check and deepen the understanding of their peers.
Autonomy	• Students are capable of checking their understanding of complex concepts on their own. They can specify open questions precisely and know where to get help in solving them. • Students can put their knowledge in relation to the contents of other lectures. • Students have developed sufficient persistence to be able to work for longer periods in a goal-oriented manner on hard problems.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** / **Bonus** / **Form** / **Description** No / 5 % / Excercises /
Examination	Oral exam
Examination duration and scale	30 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Interdisciplinary Mathematics: Specialisation II. Numerical - Modelling Training: Elective Compulsory Technomathematics: Specialisation I. Mathematics: Elective Compulsory

Course L2643: Probability Theory	
Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	SoSe
Content	• Measure and probability spaces • Integration and expectation • Types of stochastic convergence • Law of large numbers • Central limit theorem • Radon-Nikodym theorem • Conditional expectation • Martingales • Markov chains • Poisson processes
Literature	H. Bauer, Probability theory and elements of measure theory, second edition, Academic Press, 1981. A. Klenke, Probability Theory: A Comprehensive Course, second edition, Springer, 2014. G. F. Lawler, Introduction to Stochastic Processes, second edition, Chapman & Hall/CRC, 2006. A. N. Shiryaev, Probability, second edition, Springer, 1996.

Course L2644: Probability Theory	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Matthias Schulte
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2005: Human-Centric Artificial Intelligence

Courses

Title	Typ	Hrs/wk	CP
Human-Centric Artificial Intelligence (L3262)	Lecture	2	3
Human-Centric Artificial Intelligence (L3263)	Recitation Section (small)	2	3

Module Responsible	Prof. Pierre-Alexandre Murena
Admission Requirements	None
Recommended Previous Knowledge	Artificial Intelligence, and Machine Learning more specifically, is often seen as a tool to automate tasks, independently of the presence of a human. In practice, humans are everywhere in the AI process, from the data creation to the use of the learned models, not forgetting the choice and specification of the algorithms. In this course, we will explore the various impacts that the presence of a human user may have in the AI pipeline. The course will present important aspects such as explainability, or human-AI interaction.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	
Skills	
Personal Competence	
Social Competence	
Autonomy	
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	30 minutes
Assignment for the Following Curricula	Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory

Course L3262: Human-Centric Artificial Intelligence

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Pierre-Alexandre Murena
Language	EN
Cycle	SoSe
Content	Artificial Intelligence, and Machine Learning more specifically, is often seen as a tool to automate tasks, independently of the presence of a human. In practice, humans are everywhere in the AI process, from the data creation to the use of the learned models, not forgetting the choice and specification of the algorithms. In this course, we will explore the various impacts that the presence of a human user may have in the AI pipeline. The course will present important aspects such as explainability, or human-AI interaction.
Literature	

Course L3263: Human-Centric Artificial Intelligence

Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Pierre-Alexandre Murena
Language	EN
Cycle	SoSe
Content	Artificial Intelligence, and Machine Learning more specifically, is often seen as a tool to automate tasks, independently of the presence of a human. In practice, humans are everywhere in the AI process, from the data creation to the use of the learned models, not forgetting the choice and specification of the algorithms. In this course, we will explore the various impacts that the presence of a human user may have in the AI pipeline. The course will present important aspects such as explainability, or human-AI interaction.
Literature	

Module M0673: Information Theory and Coding

Courses

Title	Typ	Hrs/wk	CP
Information Theory and Coding (L0436)	Lecture	3	4
Information Theory and Coding (L0438)	Recitation Section (large)	2	2

Module Responsible	Prof. Gerhard Bauch
Admission Requirements	None
Recommended Previous Knowledge	• Mathematics 1-3 • Probability theory and random processes • Basic knowledge of communications engineering (e.g. from lecture "Fundamentals of Communications and Random Processes")
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students know the basic definitions for quantification of information in the sense of information theory. They know Shannon's source coding theorem and channel coding theorem and are able to determine theoretical limits of data compression and error-free data transmission over noisy channels. They understand the principles of source coding as well as error-detecting and error-correcting channel coding. They are familiar with the principles of decoding, in particular with modern methods of iterative decoding. They know fundamental coding schemes, their properties and decoding algorithms. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems.
Skills	The students are able to determine the limits of data compression as well as of data transmission through noisy channels and based on those limits to design basic parameters of a transmission scheme. They can estimate the parameters of an error-detecting or error-correcting channel coding scheme for achieving certain performance targets. They are able to compare the properties of basic channel coding and decoding schemes regarding error correction capabilities, decoding delay, decoding complexity and to decide for a suitable method. They are capable of implementing basic coding and decoding schemes in software.
Personal Competence	
Social Competence	The students can jointly solve specific problems.
Autonomy	The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Core Qualification: Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory

Course L0436: Information Theory and Coding

Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe

Content	• Introduction to information theory and coding
	• Definitions of information: Self information, entropy
	• Binary entropy function
	• Source coding theorem
	• Entropy of continuous random variables: Differential entropy, differential entropy of uniformly and Gaussian distributed random variables
	• Source coding
	    ◦ Principles of lossless source coding
	    ◦ Optimal source codes
	    ◦ Prefix codes, prefix-free codes, instantaneous codes
	    ◦ Morse code
	    ◦ Huffman code
	    ◦ Shannon code

  - Bounds on the average codeword length
  - Relative entropy, Kullback-Leibler distance, Kullback-Leibler divergence
  - Cross entropy
  - Lempel-Ziv algorithm
  - Lempel-Ziv-Welch (LZW) algorithm
  - Text compression and image compression using variants of the Lempel-Ziv algorithm
- Channel models
  - AWGN channel
  - Binary-input AWGN channel
  - Binary symmetric channel (BSC)
  - Relationship between AWGN channel and BSC
  - Binary error and erasure channel (BEEC)
  - Binary erasure channel (BEC)
  - Discrete memoryless channels (DMC)
- Definitions of information for multiple random variables
  - Mutual information and channel capacity
  - Entropy, conditional entropy
  - Chain rules for entropy and mutual information
- Channel coding theorem
- Channel capacity of fundamental channels: BSC, BEC, AWGN channel, binary-input AWGN channel etc.
- Power-limited vs. bandlimited transmission
- Capacity of parallel AWGN channels
  - Waterfilling
  - Examples: Multiple input multiple output (MIMO) channels, complex equivalent baseband channels, orthogonal frequency division multiplex (OFDM)
- Source-channel coding theorem, separation theorem
- Multiuser information theory
  - Multiple access channel (MAC)
  - Broadcast channel
  - Principles of multiple access, time division multiple access (TDMA), frequency division multiple access (FDMA), non-orthogonal multiple access (NOMA), hybrid multiple access
  - Achievable rate regions of TDMA and FDMA with power constraint, energy constraint, power spectral density constraint, respectively
  - Achievable rate region of the two-user and K-user multiple access channels
  - Achievable rate region of the two-user and K user broadcast channels
  - Multiuser diversity
- Channel coding
  - Principles and types of channel coding
  - Code rate, data rate, Hamming distance, minimum Hamming distance, Hamming weight, minimum Hamming weight
  - Error detecting and error correcting codes
  - Simple block codes: Repetition codes, single parity check codes, Hamming code, etc.
  - Syndrome decoding
  - Representations of binary data
  - Non-binary symbol alphabets and non-binary codes
  - Code and encoder, systematic and non-systematic encoders
  - Properties of Hamming distance and Hamming weight
  - Decoding spheres
  - Perfect codes
  - Linear codes
  - Decoding principles
    - Syndrome decoding
    - Maximum a posteriori probability (MAP) decoding and maximum likelihood (ML) decoding
    - Hard decision and soft decision decoding
    - Log-likelihood ratios (LLRs), boxplus operation
    - MAP and ML decoding using log-likelihood ratios
    - Soft-in soft-out decoders
  - Error rate performance comparison of codes in terms of SNR per info bit vs. SNR per code bit
  - Linear block codes
    - Generator matrix and parity check matrix, properties of generator matrix and parity check matrix
    - Dual codes
  - Low density parity check (LDPC) codes
    - Sparse parity check matrix
    - Tanner graphs, cycles and girth
    - Degree distributions
    - Code rate and degree distribution
    - Regular and irregular LDPC codes
    - Message passing decoding
      - Message passing decoding in binary erasure channels (BEC)
      - Systematic encoding using erasure message passing decoding
      - Message passing decoding in binary symmetric channels (BSC)
        - Extrinsic information
        - Bit-flipping decoding
        - Effects of short cycles in the Tanner graph
        - Alternative bit-flipping decoding
        - Soft decision message passing decoding: Sum product decoding

	- Bit error rate performance of LDPC codes - Repeat accumulate codes and variants of repeat accumulate codes - Message passing decoding and turbo decoding of repeat accumulate codes ◦ Convolutional codes - Encoding using shift registers - Trellis representation - Hard decision and soft decision Viterbi decoding - Bit error rate performance of convolutional codes - Asymptotic coding gain - Viterbi decoding complexity - Free distance and optimum convolutional codes - Generator polynomial description and octal description - Catastrophic convolutional codes - Non-systematic and recursive systematic convolutional (RSC) encoders - Rate compatible punctured convolutional (RCPC) codes - Hybrid automatic repeat request (HARQ) with incremental redundancy - Unequal error protection with punctured convolutional codes - Error patterns of convolutional codes ◦ Concatenated codes - Serial concatenated codes - Parallel concatenated codes, Turbo codes - Iterative decoding, turbo decoding - Bit error rate performance of turbo codes - Interleaver design for turbo codes ◦ Coded modulation - Principle of coded modulation - Achievable rates with PSK/QAM modulation - Trellis coded modulation (TCM) - Set partitioning - Ungerböck codes - Multilevel coding - Bit-interleaved coded modulation
Literature	Bossert, M.: Kanalcodierung. Oldenbourg. Friedrichs, B.: Kanalcodierung. Springer. Lin, S., Costello, D.: Error Control Coding. Prentice Hall. Roth, R.: Introduction to Coding Theory. Johnson, S.: Iterative Error Correction. Cambridge. Richardson, T., Urbanke, R.: Modern Coding Theory. Cambridge University Press. Gallager, R. G.: Information theory and reliable communication. Whiley-VCH Cover, T., Thomas, J.: Elements of information theory. Wiley.

Course L0438: Information Theory and Coding	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2080: Practical Distributed Optimization in julia

Courses

Title	Typ	Hrs/wk	CP
Practical Distributed Optimization in julia (L3449)	Lecture	2	2
Practical Distributed Optimization in julia (L3450)	Recitation Section (small)	2	4

Module Responsible	Prof. Timm Faulwasser
Admission Requirements	None
Recommended Previous Knowledge	Linear algebra, differentiation of vector-valued functions, basic programming
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Numerical optimization forms the basis of diverse technical systems and innovation. In the lecture, students learn about distributed and decentralized approaches to solving convex and non-convex optimization problems. You will be introduced to the different terminology for distributed algorithms and multi-agent systems in the context of computer science, control and optimization. Application examples from control and automation and federated learning are considered. In practical flipped classroom units, students also learn how such optimization approaches can be implemented in the Julia programming language using application examples. Specifically, the following algorithms are discussed: - Decomposition of Sequential Quadratic Programming and Interior Point methods - Augmented Lagrangian - Dual decomposition - Augmented Direction of Multipliers Methods (ADMM) - Augmented Lagrangian Inexact Newton (ALADIN) Literature - Boyd, Stephen, Neal Parikh, Eric Chu, Borja Peleato, and Jonathan Eckstein. “Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers”. Foundations and Trends® in Machine Learning 3, No. 1 (2011): 1-122 - Bertsekas, Dimitri P., and John N. Tsitsiklis. Parallel and Distributed Computation: Numerical Methods. Athena Scientific, 1997
Skills	Students are able to independently deal with multi-agent optimization issues in technical applications using mathematical methods. In particular, they are able to analyze application-related problems and transcribe them into abstract optimization problems and solve these using suitable multi-agent approaches, i.e. distributed and decentralized optimization methods. Learning objectives Students master the basics of the Julia programming language and are able to solve optimization problems in it. You have an overview of established methods for solvingconvex and non-convex optimization problems using multi-agent approaches for distributed and decentralized optimization methods.
Personal Competence	
Social Competence	The students can collaborate across boundaries of disciplines and in international teams.
Autonomy	The student can formulate questions and problems with respect to complex issues. They can program selected techniques on their own in Python.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** Yes · **Bonus** 20 % · **Form** Written elaboration · **Description**
Examination	Oral exam
Examination duration and scale	30 min
Assignment for the Following Curricula	Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory

Course L3449: Practical Distributed Optimization in julia

Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	see Module description
Literature	

Course L3450: Practical Distributed Optimization in julia	
Typ	Recitation Section (small)
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Timm Faulwasser
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2007: Probabilistic Machine Learning

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Probabilistic Machine Learning (L3268)	Lecture	2	3
Probabilistic Machine Learning (L3269)	Recitation Section (small)	2	3

Module Responsible	Prof. Pierre-Alexandre Murena
Admission Requirements	None
Recommended Previous Knowledge	Machine learning consists in inferring models to explain observed data. Managing uncertainty is essential in this process, since data may or may not have information to discriminate between several models. It is essential in various domains, such as robotics, where uncertainty about the environment has to be considered before making decisions with heavy consequences. In this course, we will explore how uncertainty can be addressed in the context of machine learning. This is done by using probabilistic models, which raise computational issues very different from those of standard ML methods. The course has two main goals: understanding the specificities of probabilistic models, and mastering the specific algorithms used to overcome the induced challenges.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	
Skills	
Personal Competence	
Social Competence	
Autonomy	
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	30 minutes
Assignment for the Following Curricula	Computational Methods and Machine Learning in Engineering: Core Qualification: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Mathematics: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory

Course L3268: Probabilistic Machine Learning

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Pierre-Alexandre Murena
Language	EN
Cycle	WiSe
Content	Machine learning consists in inferring models to explain observed data. Managing uncertainty is essential in this process, since data may or may not have information to discriminate between several models. It is essential in various domains, such as robotics, where uncertainty about the environment has to be considered before making decisions with heavy consequences. In this course, we will explore how uncertainty can be addressed in the context of machine learning. This is done by using probabilistic models, which raise computational issues very different from those of standard ML methods. The course has two main goals: understanding the specificities of probabilistic models, and mastering the specific algorithms used to overcome the induced challenges.
Literature	

Course L3269: Probabilistic Machine Learning

Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Pierre-Alexandre Murena
Language	EN
Cycle	WiSe
Content	Machine learning consists in inferring models to explain observed data. Managing uncertainty is essential in this process, since data may or may not have information to discriminate between several models. It is essential in various domains, such as robotics, where uncertainty about the environment has to be considered before making decisions with heavy consequences. In this course, we will explore how uncertainty can be addressed in the context of machine learning. This is done by using probabilistic models, which raise computational issues very different from those of standard ML methods. The course has two main goals: understanding the specificities of probabilistic models, and mastering the specific algorithms used to overcome the induced challenges.
Literature	

Module M2001: Reinforcement Learning

Courses

Title	Typ	Hrs/wk	CP
Reinforcement Learning (L3305)	Lecture	2	3
Reinforcement Learning (L3306)	Project-/problem-based Learning	2	3

Module Responsible	Dr. Manfred Eppe
Admission Requirements	None

Recommended Previous Knowledge	• Lectures at TUHH:
	  ◦ Machine Learning I and II
	  ◦ Advanced Machine Learning
	  ◦ Programming Paradigms
	• Additional recommended knowledge and prerequisites:
	  ◦ Python programming, including libraries for neural networks, deep learning and data visualization (e.g. wandb).
	  ◦ Access to a decent PC or laptop with at least mediocre performance.

Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	

Knowledge	• Foundations of Reinforcement Learning, including
	  ◦ Markov Decision Processes
	  ◦ Dynamic Programming
	  ◦ Value- and Policy Iteration
	  ◦ TD-Learning and MC-Methods
	  ◦ Policy Gradient Methods
	  ◦ Q-Learning
	  ◦ Actor-Critic Methods
	• Individual advanced competences, at least one of the following:
	  ◦ Soft Actor-Critic
	  ◦ Proximal Policy Optimization
	  ◦ Multi-Agent RL
	  ◦ Goal-Conditioned RL
	  ◦ Intrinsic Rewards and Reward Shaping
	  ◦ Hierarchical RL
	  ◦ Model-based RL

Skills	• Using a research framework for scientific questions in reinforcement learning • Data visualization and empirical evaluation of research data • Systematic validation of hypotheses • Transfer of theoretical background to practical applications
Personal Competence	
Social Competence	• Working in a team • Conflict resolution in a team • Coordination of tasks in a team
Autonomy	• Improvement of English writing and speaking • Scientific work • Presentation skills
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** Yes; **Bonus** 20 %; **Form** Presentation; **Description** Projektpräsentation
Examination	Oral exam
Examination duration and scale	30 min
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation I. Data Science & Mathematics: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory

Course L3305: Reinforcement Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Manfred Eppe, Frank Röder
Language	EN
Cycle	WiSe
Content	How can robotic and artificial agents learn to act appropriately in their environment? This course conveys foundational and advanced concepts of Reinforcement Learning to address this question. The core concepts will be taught in a lecture. In addition, there will be a strong focus on practical project work in the seminar part of this course. The practical applications will involve virtual agents in simulated 2d- or 3d environments. Prerequisites for participation are knowledge in machine learning, neural networks, stochastics, Python programming and a decent own PC or laptop with at least mediocre performance. The corresponding project/problem-based course complements the theoretical content of the lecture with a specific technical problem (mini-projects) that students solve in group work throughout the semester. The topics come from the field of robotics and are solved using 3D simulations in our institute's virtual robotics laboratory (Scilab-RL).
Literature	Reinforcement Learning: An Introduction (Sutton & Barto) Wiki and website of our virtual robotics lab (Scilab-RL): https://scilab-rl.github.io/Scilab-RL/

Course L3306: Reinforcement Learning	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Dr. Manfred Eppe, Frank Röder
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Specialization II. Computer Science

Module M1694: Security of Cyber-Physical Systems

Courses

Title	Typ	Hrs/wk	CP
Security of Cyber-Physical Systems (L2691)	Integrated Lecture	3	4
Security of Cyber-Physical Systems (L2692)	Recitation Section (small)	1	2

Module Responsible	Prof. Sibylle Fröschle
Admission Requirements	None
Recommended Previous Knowledge	IT security, programming skills, statistics
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students know and can explain - the threats posed by cyber attacks to cyber-physical systems (CPS) - concrete attacks at a technical level, e.g. on bus systems - security solutions specific to CPS with their capabilities and limitations - examples of security architectures for CPS and the requirements they guarantee - standard security engineering processes for CPS
Skills	The students are able to - identify security threats and assess the risks for a given CPS - apply attack toolkits to analyse a networked control system, and detect attacks beyond those taught in class - identify and apply security solutions suitable to the requirements - follow security engineering processes to develop a security architecture for a given CPS - recognize challenges and limitations, e.g. posed by novel types of attack
Personal Competence	
Social Competence	The students are able to - expertly discuss security risks and incidents of CPS and their mitigation in a solution-oriented fashion with experts and non-experts - foster a security culture with respect to CPS and the corresponding critical infrastructures
Autonomy	The students are able to - follow up and critically assess current developments in the security of CPS including relevant security incidents - master a new topic within the area by self-study and self-initiated interaction with experts and peers.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** No · **Bonus** 10 % · **Form** Excercises · **Description** Die Übungsaufgaben finden semesterbegleitend statt.
Examination	Written exam
Examination duration and scale	120 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory

Course L2691: Security of Cyber-Physical Systems	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	WiSe

Content	Embedded systems in energy, production, and transportation are currently undergoing a technological transition to highly networked automated cyber-physical systems (CPS). Such systems are potentially vulnerable to cyber attacks, and these can have physical impact. In this course we investigate security threats, solutions and architectures that are specific to CPS. The topics are as follows:
	
	Fundamentals and motivating examples
	
	Networked and embedded control systems
	
	    Bus system level attacks
	
	    Intruder detection systems (IDS), in particular physics-based IDS
	
	    System security architectures, including cryptographic solutions
	
	Adversarial machine learning attacks in the physical world
	
	Aspects of Location and Localization
	
	Wireless networks and infrastructures for critical applications
	
	    Communication security architectures and remaining threats
	
	    Intruder detection systems (IDS), in particular data-centric IDS
	
	    Resilience against multi-instance attacks
	
	Security Engineering of CPS: Process and Norms

Literature	Recent scientific papers and reports in the public domain.

Course L2692: Security of Cyber-Physical Systems	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0753: Software Verification

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Software Verification (L0629)	Lecture	2	3
Software Verification (L0630)	Recitation Section (small)	2	3

Module Responsible	Prof. Sibylle Schupp
Admission Requirements	None
Recommended Previous Knowledge	• Automata theory and formal languages • Computational logic • Object-oriented programming, algorithms, and data structures • Functional programming or procedural programming • Concurrency
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students apply the major verification techniques in model checking and deductive verification. They explain in formal terms syntax and semantics of the underlying logics, and assess the expressivity of different logics as well as their limitations. They classify formal properties of software systems. They find flaws in formal arguments, arising from modeling artifacts or underspecification.
Skills	Students formulate provable properties of a software system in a formal language. They develop logic-based models that properly abstract from the software under verification and, where necessary, adapt model or property. They construct proofs and property checks by hand or using tools for model checking or deductive verification, and reflect on the scope of the results. Presented with a verification problem in natural language, they select the appropriate verification technique and justify their choice.
Personal Competence	
Social Competence	Students discuss relevant topics in class. They defend their solutions orally. They communicate in English.
Autonomy	Using accompanying on-line material for self study, students can assess their level of knowledge continuously and adjust it appropriately. Working on exercise problems, they receive additional feedback. Within limits, they can set their own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of software verification. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. They can devise plans to arrive at new solutions or assess existing ones.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** Yes; **Bonus** 15 %; **Form** Excercises; **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory

Course L0629: Software Verification

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	◦ Model checking (bounded model checking, CTL, LTL) ◦ Real-time model checking (TCTL, timed automata) ◦ Deductive verification (Hoare logic) ◦ Tool support ◦ Recent developments of verification techniques and applications
Literature	• C. Baier and J-P. Katoen, Principles of Model Checking, MIT Press 2007. • M. Huth and M. Bryan, Logic in Computer Science. Modelling and Reasoning about Systems, 2nd Edition, 2004. • Selected Research Papers

Course L0630: Software Verification	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0676: Digital Communications

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Digital Communications (L0444)	Lecture	2	3
Digital Communications (L0445)	Recitation Section (large)	2	2
Laboratory Digital Communications (L0646)	Practical Course	1	1

Module Responsible	Prof. Gerhard Bauch
Admission Requirements	None
Recommended Previous Knowledge	• Mathematics 1-3 • Signals and Systems • Fundamentals of Communications and Random Processes
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students are able to understand, compare and design modern digital information transmission schemes. They are familiar with the properties of linear and non-linear digital modulation methods. They can describe distortions caused by transmission channels and design and evaluate detectors including channel estimation and equalization. They know the principles of single carrier transmission and multi-carrier transmission as well as the fundamentals of basic multiple access schemes. The students are familiar with the contents of lecture and tutorials. They can explain and apply them to new problems.
Skills	The students are able to design and analyse a digital information transmission scheme including multiple access. They are able to choose a digital modulation scheme taking into account transmission rate, required bandwidth, error probability, and further signal properties. They can design an appropriate detector including channel estimation and equalization taking into account performance and complexity properties of suboptimum solutions. They are able to set parameters of a single carrier or multi carrier transmission scheme and trade the properties of both approaches against each other.
Personal Competence	
Social Competence	The students can jointly solve specific problems.
Autonomy	The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period by solving tutorial problems, software tools, clicker system.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** Yes · **Bonus** None · **Form** Written elaboration · **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Core Qualification: Compulsory Electrical Engineering: Core Qualification: Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems: Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory

Course L0444: Digital Communications

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	• Repetition: Baseband Transmission ◦ Pulse shaping: Non-return to zero (NRZ) rectangular pulses, raised-cosine pulses, square-root raised-cosine pulses ◦ Power spectral density (psd) of baseband signals ◦ Intersymbol interference (ISI) ◦ First and second Nyquist criterion ◦ AWGN channel ◦ Matched filter ◦ Matched-filter receiver and correlation receiver ◦ Noise whitening matched filter ◦ Discrete-time AWGN channel model • Representation of bandpass signals and systems in the equivalent baseband ◦ Quadrature amplitude modulation (QAM) ◦ Equivalent baseband signal and system ◦ Analytical signal

  - Equivalent baseband random process, equivalent baseband white Gaussian noise process
  - Equivalent baseband AWGN channel
  - Equivalent baseband channel model with frequency-offset and phase noise
  - Equivalent baseband Rayleigh fading and Rice fading channel models
  - Equivalent baseband frequency-selective channel model
  - Discrete memoryless channels (DMC)
- Bandpass transmission via carrier modulation
  - Amplitude modulation, frequency modulation, phase modulation
  - Linear digital modulation methods
    - On-off keying, M-ary amplitude shift keying (M-ASK), M-ary phase shift keying (M-PSK), M-ary quadrature amplitude modulation (M-QAM), offset-QPSK
    - Signal space representation of transmit signal constellations and signals
    - Energy of linear digital modulated signals, average energy per symbol
    - Power spectral density of linear digital modulated signals
    - Bandwidth efficiency
    - Correlation coefficient of elementary signals
    - Error probabilities of linear digital modulation methods
      - Error functions
      - Gray mapping and natural mapping
      - Bit error probabilities, symbol error probabilities, pairwise symbol error probabilities
      - Euclidean distance and Hamming distance
      - Exact and approximate computation of error probabilities
      - Performance comparison of modulation schemes in terms of per bit SNR vs. per symbol SNR
    - Hierarchical modulation, multilevel modulation
    - Effects of carrier phase offset and carrier frequency offset
    - Differential modulation
      - M-ary differential phase shift keying (M-PSK)
      - Coherent and non-coherent detection of DPSK
      - p/M-differential phase shift keying (p/M-DPSK)
      - Differential amplitude and phase shift keying (DAPSK)
  - Non-linear digital modulation methods
    - Frequency shift keying (FSK)
    - Modulation index
    - Minimum shift keying (MSK)
      - Offset-QPSK representation of MSK
      - MSK with differential precoding and rotation
      - Bit error probabilities of MSK
      - Gaussian minimum shift keying (GMSK)
      - Power spectral density of MSK and GMSK
    - Continuous phase modulation (CPM)
      - General description of CPM signals
      - Frequency pulses and phase pulses
    - Coherent and non-coherent detection of FSK
  - Performance comparison of linear and non-linear digital modulation methods
- Frequency-selective channels, ISI channels
  - Intersymbol interference and frequency-selectivity
  - RMS delay spread
  - Narrowband and broadband channels
  - Equivalent baseband transmission model for frequency-selective channels
  - Receive filter design
- Equalization
  - Symbol-spaced and fractionally-spaced equalizers
  - Inverse system
  - Non-recursive linear equalizers
    - Linear zero-forcing (ZF) equalizer
    - Linear minimum mean squared error (MMSE) equalizer
  - Non-linear equalization:
    - Decision feedback equalizer (DFE)
    - Tomlinson-Harashima precoding
  - Maximum a posteriori probability (MAP) and maximum likelihood equalizer, Viterbi algorithm
- Single-carrier vs. multi-carrier transmission
- Multi-carrier transmission
  - General multicarrier transmission
  - Orthogonal frequency division multiplex (OFDM)
    - OFDM implementation using the Fast Fourier Transform (FFT)
    - Cyclic guard interval
    - Power spectral density of OFDM
    - Peak-to-average power ratio (PAPR)
- Multiple access
  - Principles of time division multiple access (TDMA), frequency division multiple access (FDMA), code division multiple access (CDMA), non-orthogonal multiple access (NOMA), hybrid multiple access
- Spread spectrum communications
  - Direct sequence spread spectrum communications
  - Frequency hopping
  - Protection against eavesdropping

	- Protection against narrowband jammers
	- Short vs. long spreading codes
	- Direct sequence spread spectrum communications in frequency-selective channels
	  - Rake receiver
	- Code division multiple access (CDMA)
	  - Design criteria of spreading sequences, autocorrelation function and crosscorrelation function of spreading sequences
	  - Intersymbol interference (ISI) and multiple access interference (MAI)
	  - Pseudo noise (PN) sequences, maximum length sequences (m-sequences), Gold codes, Walsh-Hadamard codes, orthogonal variable spreading factor (OVSF) codes
	  - Multicode transmission
	  - CDMA in uplink and downlink of a wireless communications system
	  - Single-user detection vs. multi-user detection

Literature	K. Kammeyer: Nachrichtenübertragung, Teubner P.A. Höher: Grundlagen der digitalen Informationsübertragung, Teubner. J.G. Proakis, M. Salehi: Digital Communications. McGraw-Hill. S. Haykin: Communication Systems. Wiley R.G. Gallager: Principles of Digital Communication. Cambridge A. Goldsmith: Wireless Communication. Cambridge. D. Tse, P. Viswanath: Fundamentals of Wireless Communication. Cambridge.

Course L0445: Digital Communications	
Typ	Recitation Section (large)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0646: Laboratory Digital Communications	
Typ	Practical Course
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Gerhard Bauch
Language	EN
Cycle	WiSe
Content	- DSL transmission - Random processes - Digital data transmission
Literature	K. Kammeyer: Nachrichtenübertragung, Teubner P.A. Höher: Grundlagen der digitalen Informationsübertragung, Teubner. J.G. Proakis, M. Salehi: Digital Communications. McGraw-Hill. S. Haykin: Communication Systems. Wiley R.G. Gallager: Principles of Digital Communication. Cambridge A. Goldsmith: Wireless Communication. Cambridge. D. Tse, P. Viswanath: Fundamentals of Wireless Communication. Cambridge.

Module M1598: Image Processing

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Image Processing (L2443)	Lecture	2	4
Image Processing (L2444)	Recitation Section (small)	2	2

Module Responsible	Prof. Tobias Knopp
Admission Requirements	None
Recommended Previous Knowledge	Signal and Systems
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students know about • visual perception • multidimensional signal processing • sampling and sampling theorem • filtering • image enhancement • edge detection • multi-resolution procedures: Gauss and Laplace pyramid, wavelets • image compression • image segmentation • morphological image processing
Skills	The students can • analyze, process, and improve multidimensional image data • implement simple compression algorithms • design custom filters for specific applications
Personal Competence	
Social Competence	Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem.
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Data Science: Specialisation I. Mathematics/Computer Science: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Signal Processing: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L2443: Image Processing	
Typ	Lecture
Hrs/wk	2
CP	4
Workload in Hours	Independent Study Time 92, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Visual perception • Multidimensional signal processing • Sampling and sampling theorem • Filtering • Image enhancement • Edge detection • Multi-resolution procedures: Gauss and Laplace pyramid, wavelets • Image Compression • Segmentation • Morphological image processing
Literature	Bredies/Lorenz, Mathematische Bildverarbeitung, Vieweg, 2011 Pratt, Digital Image Processing, Wiley, 2001 Bernd Jähne: Digitale Bildverarbeitung - Springer, Berlin 2005

Course L2444: Image Processing	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1774: Advanced Internet Computing

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Advanced Internet Computing (L2916)	Lecture	2	3
Advanced Internet Computing (L2917)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Stefan Schulte
Admission Requirements	None
Recommended Previous Knowledge	Good programming skills are necessary. Previous knowledge in the field of distributed systems is helpful.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	After successful completion of the course, students are able to: • Describe basic concepts of cloud computing and blockchain technologies • Discuss and assess critical aspects of cloud computing and blockchain technologies • Select and apply cloud and blockchain technologies for particular application areas • Design and develop practical solutions based on cloud and blockchain software
Skills	The students acquire the ability to model Internet-based distributed systems and to work with these systems. This comprises especially the ability to select and utilize fitting technologies for different application areas. Furthermore, students are able to critically assess the chosen technologies.
Personal Competence	
Social Competence	Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem.
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Group project incl. presentation (50 %), written exam (60 min, 50 %)
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory

Course L2916: Advanced Internet Computing

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This lecture discusses modern Internet-based distributed systems in two blocks: (i) cloud computing and (ii) blockchain technologies. The following topics will be covered in the single lectures: • Cloud Service and Deployment Models • Virtualization • Further Aspects of Cloud Computing • Edge and Fog Computing • Blockchain technologies • Consensus
Literature	Lecture notes as well as current literature announced in the lecture.

Course L2917: Advanced Internet Computing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Stefan Schulte
Language	EN
Cycle	SoSe
Content	This project-/problem-oriented part of the module augments the theoretical content of the lecture by a concrete technical problem, which needs to be solved by the students in group work during the semester. Possible topics are (blockchain-based) sensor data integration and blockchain-based voting.
Literature	Lecture notes as well as current literature announced in the lecture.

Module M1301: Software Testing

Courses

Title	Typ	Hrs/wk	CP
Software Testing (L1791)	Lecture	2	3
Software Testing (L1792)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Sibylle Schupp
Admission Requirements	None
Recommended Previous Knowledge	• Software Engineering • Higher Programming Languages • Object-Oriented Programming • Algorithms and Data Structures • Experience with (Small) Software Projects • Statistics
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students explain the different phases of testing, describe fundamental techniques of different types of testing, and paraphrase the basic principles of the corresponding test process. They give examples of software development scenarios and the corresponding test type and technique. They explain algorithms used for particular testing techniques and describe possible advantages and limitations.
Skills	Students identify the appropriate testing type and technique for a given problem. They adapt and execute respective algorithms to execute a concrete test technique properly. They interpret testing results and execute corresponding steps for proper re-test scenarios. They write and analyze test specifications. They apply bug finding techniques for non-trivial problems.
Personal Competence	
Social Competence	Students discuss relevant topics in class. They defend their solutions orally. They communicate in English.
Autonomy	Students can assess their level of knowledge continuously and adjust it appropriately, based on feedback and on self-guided studies. Within limits, they can own learning goals. Upon successful completion, students can identify and precisely formulate new problems in academic or applied research in the field of testing. Within this field, they can conduct independent studies to acquire the necessary competencies and compile their findings in academic reports. T devise plans to arrive at new solutions or assess existing ones
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Software
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory

Course L1791: Software Testing	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	• Fundamentals of software testing • Model-based testing • Test automation • Criteria-based testing
Literature	• P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2016. • Selected research papers

Course L1792: Software Testing	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	SoSe
Content	• Fundamentals of software testing • Model-based testing • Test automation • Criteria-based testing
Literature	• M. Pezze and M. Young, Software Testing and Analysis, John Wiley 2008. • P. Ammann and J. Offutt, "Introduction to Software Testing", 2nd edition 2015.

Module M1810: Autonomous Cyber-Physical Systems

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Autonomous Cyber-Physical Systems (L3000)	Lecture	2	3
Autonomous Cyber-Physical Systems (L3001)	Project-/problem-based Learning	3	3

Module Responsible	Prof. Bernd-Christian Renner
Admission Requirements	None
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C/C++ language (e.g., module: Procedural Programming for Computer Scientists) • Basic knowledge in software engineering • Basic knowledge in wired and wireless communication protocols • Principal understanding of simple electronic circuits
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	This master's course centers on the advanced Internet of Things (IoT) systems, spotlighting energy sustainability and Real-Time Operating Systems (RTOS) within cyber-physical frameworks. Key focus areas include the exploration of IoT system architecture, especially leveraging RTOS for real-time responsiveness, and energy-efficient technologies for sustainable operations. Upon completion, students will be capable of: • Outlining the structure and operational principles of IoT systems and their significance in modern cyber-physical environments. • Differentiating between RTOS and conventional operating systems in the context of IoT applications. • Evaluating energy harvesting (specifically solar technologies) and storage systems for their applicability and efficiency in IoT devices • Knowing different wireless technologies (e.g., Bluetooth LE, LoRaWAN, Zigbee) and networking protocols (TCP/IP, MQTT) for IoT connectivity and functionality, and pinpointing their dis-/advantages.
Skills	Students will acquire practical skills to: • Describe, plan, develop and optimize IoT applications using RTOS, focusing on multitasking, efficient memory, and power management. • Select and deploy suitable wireless and networking technologies considering device requirements and operational contexts. • Integrate energy harvesting and storage technologies into IoT solutions, enhancing their sustainability and autonomy. • Conduct comprehensive planning and evaluation of IoT projects, incorporating scientific methodologies for real-world applications.
Personal Competence	
Social Competence	• Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • After completing the module, the students are able to work on similar tasks alone or in a group and to present the results in a suitable way.
Autonomy	After completing the module, the students are able to independently work on sub-areas of the subject using specialist literature, to summarize and present the knowledge they have acquired and to link it to the content of other courses.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** No; **Bonus** 10 %; **Form** Attestation; **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Mechatronics: Technical Complementary Course: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L3000: Autonomous Cyber-Physical Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	
Literature	

Course L3001: Autonomous Cyber-Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1682: Secure Software Engineering

Courses

Title	Typ	Hrs/wk	CP
Secure Software Engineering (L2667)	Lecture	2	3
Secure Software Engineering (L2668)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Riccardo Scandariato
Admission Requirements	None
Recommended Previous Knowledge	Familiarity with basic software engineering concepts (e.g., requirements, design) and basic security concepts (e.g., confidentiality, integrity, availability)
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students can: • Elicit security requirements in a software project • Model and document security measures in a software design • Use threat and risk analysis techniques • Understand how security code reviews are performed • Understand the core definitions of concepts related to privacy • Understand privacy enhancing technologies
Skills	Select appropriate security assurance techniques to be used in a security assurance program
Personal Competence	
Social Competence	None
Autonomy	Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, techical standards, and white papers.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** / **Bonus** / **Form** / **Description** No / 5 % / Subject theoretical and practical work / Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam
Examination duration and scale	120 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Software and Signal Processing: Elective Compulsory

Course L2667: Secure Software Engineering	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	• Secure software development processes and maturity models • Techniques to define security requirements • Techniques to create, document and analyse the design of secure applications • Threat and risk analysis techniques • Security code reviews • Program repair techniques for security vulnerabilities • Privacy engineering
Literature	Charles Haley, Robin Laney, Jonathan Moffett, Bashar Nuseibeh, 2008, Security Requirements Engineering: A Framework for Representation and Analysis, Transactions on Software Engineering Sindre, G. and Opdahl, A.L., 2005. Eliciting security requirements with misuse cases. Requirements engineering, 10(1) Fontaine, P.J., Van Lamsweerde, A., Letier, E. and Darimont, R., 2001. Goal-oriented elaboration of security requirements. Joanna C. S. Santos, Katy Tarrit, Mehdi Mirakhorli, 2017, A Catalog of Security Architecture Weaknesses, IEEE International Conference on Software Architecture Workshops (ICSAW) Jürjens, J., 2002, UMLsec: Extending UML for secure systems development, International Conference on The Unified Modeling Language Lund, M.S., Solhaug, B. and Stølen, K., 2011. A guided tour of the CORAS method. In Model-Driven Risk Analysis, Springer Pratyusa Manadhata, Jeannette Wing, 2021, An Attack Surface Metric, IEEE Transactions on Software Engineering, 37(3) Howard, M.A., 2006. A process for performing security code reviews. IEEE Security & privacy, 4(4) Jaap-Henk Hoepman, 2022, Privacy Design Strategies (The Little Blue Book) Diaz, C. and Gürses, S., 2012. Understanding the landscape of privacy technologies. Proceedings of the information security summit, 12

Course L2668: Secure Software Engineering	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1773: Cybersecurity Data Science

Courses

Title	Typ	Hrs/wk	CP
Cybersecurity Data Science (L2914)	Lecture	2	3
Cybersecurity Data Science (L2915)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Riccardo Scandariato
Admission Requirements	None
Recommended Previous Knowledge	Basic knowledge of probabilities and statistics. Familiarity with object oriented programming.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students can: • Apply data science methods to the resolution of complex cybersecurity problems. • Use of data science methods to quantify risks and optimize cybersecurity operations. • Identify strengths and limitations of state-of-the-art methods. • Select the performance indicators of data-oriented cybersecurity solutions. • Understand cybersecurity threats in data science methods.
Skills	Implement and evaluate data-driven models for the identification, treatment, and mitigation of cybersecurity risks
Personal Competence	
Social Competence	None
Autonomy	Students can apply the knowledge acquired throughout the course to the resolution of industrial case studies. Students should also be capable to acquire new knowledge independently from academic publications, techical standards, and white papers.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** No; **Bonus** 5 %; **Form** Subject theoretical and practical work; **Description** Gruppenarbeit mit aktuellen Technologien aus dem Bereich Sicherheit
Examination	Written exam
Examination duration and scale	120 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory

Course L2914: Cybersecurity Data Science

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	Theoretical Foundations: • Introduction to data science • Supervised and unsupervised learning • Data science methods (e.g., clustering, decision trees, artificial neural networks) • Performance indicators • Other data science methods for cybersecurity (e.g., experiments, surveys) Cybersecutrity Applications: • Spam/Phishing detection • Intrusion detection • Vulnerability/malware prediction • Adversarial machine learning
Literature	Sarker, I.H., Kayes, A.S.M., Badsha, S., Alqahtani, H., Watters, P. and Ng, A., 2020. Cybersecurity data science: an overview from machine learning perspective. Journal of Big data, 7(1) Torres, J.M., Comesaña, C.I. and Garcia-Nieto, P.J., 2019. Machine learning techniques applied to cybersecurity. International Journal of Machine Learning and Cybernetics, 10(10) Arp, D., Quiring, E., Pendlebury, F., Warnecke, A., Pierazzi, F., Wressnegger, C., Cavallaro, L. and Rieck, K., Dos and Don'ts of Machine Learning in Computer Security.

Course L2915: Cybersecurity Data Science	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Riccardo Scandariato
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M0924: Software for Embedded Systems

Courses

Title	Typ	Hrs/wk	CP
Software for Embdedded Systems (L1069)	Lecture	2	3
Software for Embdedded Systems (L1070)	Project-/problem-based Learning	3	3

Module Responsible	Prof. Bernd-Christian Renner
Admission Requirements	None
Recommended Previous Knowledge	• Very good knowledge and practical experience in programming in the C language and its compilation process • Basic knowledge in software engineering • Basic understanding of assembly language • Basic knowledge of electrical engineering
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Students know the contents of the module and can reproduce and link them. • They are able to describe the usage and advantages of event-based programming using interrupts. • Students can explain the basic principles and procedures for creating software for embedded systems. • They know the components and functions of a concrete microcontroller. • The participants can explain requirements of real time systems. • They know at least three scheduling algorithms for real time operating systems including their pros and cons. • They know different memories for embedded systems and their use cases, including advantages and disadvantages. • They understand race conditions, their causes and consequences. • They are familiar with at least two bootloader concepts and can evaluate their suitability in different contexts. • They know mechanisms for optimizing the power consumption of microcontrollers with sleep modes.
Skills	• Students design and write hardware-oriented software modules in a high-level language (C or similar) for an embedded system based on a specific microcontroller. • They can configure and interact with peripherals (timer, ADC, EEPROM), including bit-level register operations, interrupt-based processing, and program flow. • They implement and use a (preemptive) scheduler for an embedded system. • They learn to write independent, reusable software components. They build a complex software system (library and applications) throught the semester. • They learn how to model event-based software for embedded systems with UML state diagrams and finite state machines and implement a complex system that they modeled. • They can identify race conditions in unknown software and develop strategies to work around them.
Personal Competence	
Social Competence	• Students are able to work goal-oriented in small mixed groups. • They learn and broaden their teamwork abilities. • They learn to define and split tasks within the team. • They practice to present their software, the selection of algorithms and their design choices.
Autonomy	Students are able • to solve assignments related to this lecture independently with instructional direction. • to design, implement, and test software components for an embedded system independently based on a textual description. • to read and understand data sheets and manuals of electronic components (such as micro-controllers and sensors). • to independently transfer the concepts known from the lecture to the exercises and to select algorithms based on given criteria.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** No, **Bonus** 10 %, **Form** Attestation, **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Microelectronics and Microsystems: Specialisation Embedded Systems: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L1069: Software for Embdedded Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	• General-Purpose Processors • Programming the Atmel AVR • Interrupts and Race Conditions • C for Embedded Systems • Standard Single Purpose Processors: Peripherals • Finite-State Machines • Memory • Operating Systems for Embedded Systems • Real-Time Embedded Systems • Boot loader and Power Management
Literature	1. Embedded System Design, F. Vahid and T. Givargis, John Wiley 2. Programming Embedded Systems: With C and Gnu Development Tools, M. Barr and A. Massa, O'Reilly 3. C und C++ für Embedded Systems, F. Bollow, M. Homann, K. Köhn, MITP 4. The Art of Designing Embedded Systems, J. Ganssle, Newnses 5. Mikrocomputertechnik mit Controllern der Atmel AVR-RISC-Familie, G. Schmitt, Oldenbourg 6. Making Embedded Systems: Design Patterns for Great Software, E. White, O'Reilly

Course L1070: Software for Embdedded Systems	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Bernd-Christian Renner
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1794: Applied Cryptography

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Applied Cryptography (L2954)	Integrated Lecture	3	4
Applied Cryptography (L2955)	Recitation Section (small)	1	2

Module Responsible	Prof. Sibylle Fröschle
Admission Requirements	None
Recommended Previous Knowledge	Basics in IT security including cryptography, basics in theoretical computer science including computability and logic
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students know and can explain - the core cryptographic primitives and constructions in the rigorous treatment of modern cryptography - cryptographic protocols for authentication and key establishment in the modern rigorous treatment - advanced key management techniques using trust anchors such as hardware security modules - the impact of quantum computers on cryptography and the advance towards post-quantum cryptography
Skills	The students are able to - understand subtle attacks on inadequately applied cryptography and how to avoid them - follow security standards to develop secure cryptographic applications - systematically analyze and solve complex problems in the area of cryptographic protocols and key management - recognize novel challenges and limitations, e.g. posed by novel types of attacks
Personal Competence	
Social Competence	The students are able to - expertly discuss when cryptographic solutions are necessary and what it takes to deploy them with experts and non-experts - champion the approach of modern cryptography with its focus on precise definitions and assumptions as essential for security
Autonomy	The students are able to - follow up and critically assess current developments in cryptography and cryptographic applications including the challenge of post-quantum cryptography - master a new topic within the area by self-study and self-initiated interaction with experts and peers
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** / **Bonus** / **Form** / **Description** No / 10 % / Excercises / Die Übungsaufgaben finden semesterbegleitend statt
Examination	Written exam
Examination duration and scale	120 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems, Focus Networks: Elective Compulsory

Course L2954: Applied Cryptography	
Typ	Integrated Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	Cryptography plays a key role in securing the digital world we live in today. This course provides you with a deep understanding of symmetric and public-key cryptography based on the modern approach to cryptography. The modern approach requires that cryptographic constructions are provably secure relative to precise definitions and plausible assumptions. The assumptions typically stem from computability theory and algorithmic number theory. Moreover, you will get to know how cryptography can be applied to solve security problems such as secure communication over an open channel such as the Internet, and we will cover advanced aspects of practical deployment such as key management anchored in hardware security modules. We will see how the theme of precise definition and provable security carries forth in these areas, and are a must for today's security applications. Finally, we discuss the impact of the advent of quantum computers on cryptography and its application, and the current state of post-quantum cryptography.
Literature	Introduction to Modern Cryptography, Third Edition, Jonathan Katz and Jehuda Lindell, Chapman & Hall/CRC, 2021

Course L2955: Applied Cryptography	
Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Fröschle
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1685: Selected Aspects in Computer Science

Courses

Title	Typ	Hrs/wk	CP
Selected Aspects in Computer Science (L2672)	Lecture	3	4
Selected Aspects in Computer Science (L2673)	Recitation Section (small)	1	2

Module Responsible	Prof. Görschwin Fey
Admission Requirements	None
Recommended Previous Knowledge	
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	
Skills	
Personal Competence	
Social Competence	
Autonomy	
Workload in Hours	Depends on choice of courses
Credit points	6
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Information and Communication Systems: Specialisation Secure and Dependable IT Systems: Elective Compulsory

Course L2672: Selected Aspects in Computer Science

Typ	Lecture
Hrs/wk	3
CP	4
Workload in Hours	Independent Study Time 78, Study Time in Lecture 42
Examination Form	Klausur
Examination duration and scale	90 min
Lecturer	Dozenten des SD E
Language	DE/EN
Cycle	WiSe/SoSe
Content	
Literature	

Course L2673: Selected Aspects in Computer Science

Typ	Recitation Section (small)
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Examination Form	Fachtheoretisch-fachpraktische Arbeit
Examination duration and scale	x
Lecturer	Dozenten des SD E
Language	DE/EN
Cycle	WiSe/SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2137: TinyML: Embedded Machine Learning

Courses

Title	Typ	Hrs/wk	CP
TinyML: Embedded Machine Learning (L3424)	Lecture	2	3
TinyML: Embedded Machine Learning (L3425)	Recitation Section (small)	2	3

Module Responsible	Prof. Olaf Landsiedel
Admission Requirements	None
Recommended Previous Knowledge	A student should have taken courses on Internet of Things or embedded systems, such as the course "Internet of Things". Further, we expect students to have taken a course on machine learning and especially neural networks (including knowledge of backpropagation and experience in TensorFlow, PyTorch or similar platforms). Moreover, C programming skills are expected, as we will be using embedded IoT devices.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	- Understanding of the intersection of machine learning and embedded IoT devices. - Familiarity with cutting-edge literature and frameworks in the field of TinyML. - Knowledge of efficient inference techniques, including model compression, pruning, quantization, neural architecture search, and distillation. - Understanding of efficient training techniques, such as distributed training, gradient compression, and on-device transfer learning. - Understanding of how to train and deploy models on microcontrollers.
Skills	- Practical experience through hands-on project assignments involving embedded AI systems and applications. - Ability to conceive and develop a novel TinyML application running on a microcontroller (MCU). - Application of learned concepts to real-world problems through experimentation and analysis. - C programming skills for use with embedded IoT devices.
Personal Competence	
Social Competence	hands-on project assignments promote collaboration and teamwork
Autonomy	- Ability to independently conceive, design, and develop embedded AI applications. - Capability to understand and tackle fundamental issues in designing methods for embedded AI systems and applications. - Self-driven exploration of cutting-edge literature and frameworks in TinyML to advance understanding and application of the concepts.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** Yes; **Bonus** None; **Form** Attestation; **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory

Course L3424: TinyML: Embedded Machine Learning	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	SoSe
Content	Abstract: This course is a course at the intersection of Machine Learning and Embedded IoT Devices. The pervasiveness of ultra-low-power embedded devices, coupled with the introduction of embedded machine learning frameworks, will enable the mass proliferation of AI-powered IoT devices. The rapid growth of machine learning and how easy it is to use machine learning platforms make it an important subject for computer science students to learn about today. Objectives: At the end of the course, students will have learned the following: Practical experience through hands-on project assignments Gained familiarity with cutting-edge literature and frameworks in the field of TinyML Know how to train and deploy models on microcontrollers Conceived and developed a (novel) TinyML application running on a MCU Contents: This course is a deep dive into efficient machine learning techniques that enable powerful deep learning applications on resource-constrained devices. Topics covered • Brief recap / introduction to ML and IoT • efficient inference techniques, including model compression, pruning, quantization, neural architecture search, and distillation; • and efficient training techniques, including distributed training, gradient compression and on-device transfer learning; Our lectures provide students with the required fundamentals, and exercise and projects give students a hands-on experience in developing embedded AI systems & applications and exploring their real-world challenges. This course offers learning experiences that involve hands-on experimentation and analysis as they reinforce student understanding of concepts and their application to real-world problems. Overall, this course provides the students the ability to understand fundamental issues in the design of methods for embedded AI systems and applications. Additional prerequisites: A student should have taken courses on Internet of Things or embedded systems, such as the course "Internet Things and Wireless Networks" at CAU. Further, we expect students to have taken a course on machine learning and especially neural networks (including knowledge of backpropagation and experience in TensorFlow, PyTorch or similar platforms). Moreover, C programming skills are expected, as we will be using embedded IoT devices.
Literature	1.

Course L3425: TinyML: Embedded Machine Learning	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M2090: Quantum Mechanics

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Quantum Mechanics (L3363)	Lecture	2	3
Quantum Mechanics (L3364)	Recitation Section (large)	1	1
Quantum Mechanics (L3382)	Recitation Section (small)	2	2

Module Responsible	Prof. Martin Kliesch
Admission Requirements	None
Recommended Previous Knowledge	• Knowledge of mathematics is required, including linear algebra, vector calculus, complex numbers, the Fourier transform, and wave equations. • Some Knowledge in physics is helpful: wave phenomena, harmonic oscillators (classical), and electromagnetism.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	This module is an introduction to quantum mechanics on the master level. The students learn basic paradigmatic models of quantum mechanics and acquire the ability to describe and explain the basic terms and principles. They can distinguish similarities and differences between quantum mechanics and classical physics and know in which situations quantum mechanical phenomena may be expected. An emphasis will be placed on conceptual and mathematical aspects.
Skills	After completing this module, students can • reproduce the knowledge taught in the course, • reproduce simpler proofs of the course and reproduce the ideas of the more complicated ones, • establish connections between the concepts taught and • apply the learned knowledge to concrete problems.
Personal Competence	
Social Competence	After completing this module, students are expected to be able to work on subject-specific tasks alone or in a group and to present the results appropriately.
Autonomy	After completing this module, students are able to work out sub-areas of the subject independently using textbooks and other literature to summarize and present the acquired knowledge and link it to the contents of other courses.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** No · **Bonus** 15 % · **Form** Excercises · **Description**
Examination	Written exam
Examination duration and scale	120 min
Assignment for the Following Curricula	Computer Science: Specialisation III. Mathematics: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Microelectronics and Microsystems: Core Qualification: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Robotics and Computer Science: Elective Compulsory

Course L3363: Quantum Mechanics	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	WiSe
Content	• Overview of experiments (Stern-Gerlach, black body radiation, photoelectric effect, double slit experiment) • Wave equations & dispersion relations • Schrödinger equation, wave packets, tunnelling effect • Postulates of quantum mechanics (pure states) • Harmonic oscillator • Heisenberg uncertainty relation • Hydrogen Atom • Composite quantum systems • Spin & coupling to magnetic fields • Outlook: Fermions & Bosons
Literature	• Sakurai and Napolitano, Modern Quantum Mechanics (3rd ed., 2020) • Nielsen and Chuang, Quantum Computation and Quantum Information

Course L3364: Quantum Mechanics	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L3382: Quantum Mechanics	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Martin Kliesch
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M2138: Internet of Things

Courses

Title	Typ	Hrs/wk	CP
Internet of Things (L3426)	Lecture	2	3
Internet of Things (L3427)	Recitation Section (small)	2	3

Module Responsible	Prof. Olaf Landsiedel
Admission Requirements	None
Recommended Previous Knowledge	
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	- Understanding of the foundational concepts and challenges of wireless networking and the Internet of Things (IoT). - Knowledge of low-power wireless communication technologies, such as NFC, BLE, and 802.15.4. - Understanding of IoT networking protocols like RPL, TSCH, and Thread. - Knowledge of operating systems for IoT and security aspects in IoT. - Understanding of IoT applications and the mechanisms they use to provide services. - Ability to analyze the challenges and requirements of different approaches in the IoT field. - Ability to compare and summarize the strengths and weaknesses of individual IoT mechanisms.
Skills	- Development and evaluation of small-scale wireless networks and IoT systems & applications. - Practical application of fundamental mechanisms introduced in the lectures. - Demonstration of software developments considering unreliable wireless links and resource constraints (e.g., battery-driven).
Personal Competence	
Social Competence	teamwork and collaborative project work through the "hands-on experience in developing IoT systems & applications."
Autonomy	- Ability to independently analyze and evaluate existing and new methods for IoT system design. - Application of learned concepts to real-world problems. - Independently developing solutions for challenges in the design and implementation of IoT systems, especially in terms of resource-constrained and low-power networks.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** Yes; **Bonus** None; **Form** Attestation; **Description**
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation I. Computer and Software Engineering: Elective Compulsory Data Science: Specialisation II. Computer Science: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory

Course L3426: Internet of Things	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	WiSe
Content	Abstract: This course will introduce and discuss the underlying concepts and mechanisms that drive wireless networks and the Internet of Things (IoT). The lectures provide you with the required knowledge, and the labs give you a hands-on experience in developing networking and applications in the Internet of Things and exploring their real-world challenges. Objectives: 1. Knowledge and understanding: Show knowledge of basic concepts and challenges of wireless networking and the Internet of Things, low-power wireless communication, NFC, BLE and 802.15.4), IoT networking (RPL, TSCH, and Thread), Operating Systems for IoT, Security in IoT, and IoT Applications. Describe applications of the Internet of Things and the mechanisms these use to provide their services. Discuss and analyze the challenges and requirements that the different approaches have. Compare and summarize the strength and weaknesses associated with the individual mechanisms. 2. Skills and abilities: Develop and evaluate small-scale wireless networks and IoT systems & applications using fundamental mechanisms introduced in the lectures. Demonstration of these software developments in advanced settings including unreliable wireless links and resource constraints (e.g., battery driven). 3. Judgment and approach: Describe and analyze existing and new methods for IoT systems and application design. In particular, the system's ability for low-power wireless networking and operation under strong resource constraints. Contents: Aim: What is the Internet of Things and its applications? How can we build reliable and resource efficient IoT systems and applications? How do its devices in the Internet of Things communicate? How do wireless networks work? These are a few questions that this course addresses. The goal of the courses is to understand the design of the Internet of Things and discuss the underlying principles and mechanisms. Our lectures provide you with the required fundamentals, and our labs give you a hands-on experience in developing IoT systems and applications and exploring their real-world challenges. Content: We begin the course with an introduction to basic concepts of IoT Systems and Applications and the challenges they pose. We continue with the main course content and focus on: • Wireless networking • low-power wireless communication (NFC, BLE and 802.15.4) • IoT networking (RPL, TSCH, and Thread) • web-technologies for IoT such as COAP • Operating Systems for IoT • Security in IoT • Selected IoT Applications Our lectures provide students with the required fundamentals, and exercise and projects give students a hands-on experience in developing IoT systems & applications and exploring their real-world challenges. This course offers learning experiences that involve hands-on experimentation and analysis as they reinforce student understanding of concepts and their application to real-world problems. Overall, this course provides the students the ability to understand fundamental issues in the design of methods for IoT systems and applications.
Literature	

Course L3427: Internet of Things	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Olaf Landsiedel
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Specialization III. Applications

Module M1881: Digital Health

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Digital Health (L3099)	Lecture	3	3
Digital Health Seminar (L3100)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Moritz Göldner
Admission Requirements	None
Recommended Previous Knowledge	none
Educational Objectives	After taking part successfully, students have reached the following learning results

Professional Competence

Knowledge

- Comprehensive understanding of the digital health landscape: Students will acquire knowledge about various aspects of digital health, including current trends, business models, and their impacts on the German healthcare system. They will develop a holistic understanding of the digital transformation in the healthcare sector.
- Knowledge of technologies and applications: Students will gain insights into various technologies and applications in the field of digital health, such as mHealth, Digital Health Applications (DiGA), Digital Patient Records (DiPA), and telemedicine. They will learn how these technologies function and the potential benefits they can offer.
- Understanding of physician and patient perspectives: Students will develop an understanding of the different perspectives of physicians and patients regarding digital health. They will recognize the opportunities and challenges from both viewpoints and incorporate them into their decision-making and practice.
- Knowledge of interoperability and data management: Students will comprehend the significance of interoperability and effective data management in the context of digital health to enhance the quality of digital healthcare provision.
- Knowledge of big data and AI in healthcare: Students will become familiar with the transformative role of big data and artificial intelligence in healthcare. They will explore practical applications of these technologies and consider ethical issues related to their usage.

Skills

Through engaging in paper presentations, group work, case studies, and practical exercises, students will cultivate the skills to apply their knowledge, analyze information, and devise solutions for real-world issues in the realm of digital health. Additionally, they will enhance their presentation abilities and their aptitude for collaborating within interdisciplinary teams.

Personal Competence

Social Competence

During the lecture series on "Digital Health," students acquire various social competencies that enable them to thrive in the digital health domain and collaborate effectively with other professionals. These competencies include:

- Teamwork: Students are encouraged to collaborate with their peers in group assignments and case studies. They learn to work effectively in interdisciplinary teams, solving complex problems and developing innovative approaches. This cultivates their communication and cooperation skills.
- Presentation skills: Through paper presentations and other formats, students are guided in presenting their findings and research results to their classmates. This enhances their ability to deliver content clearly and convincingly, effectively communicating their ideas.
- Discussion skills: The lecture series promotes active discussions and the exchange of diverse perspectives. Students learn to articulate their opinions and arguments, consider alternative viewpoints, and engage in constructive discussions. This fosters their critical thinking and collaboration abilities within an academic environment.
- Empathy and patient-centered approach: Exploring medical and patient perspectives on digital health enhances students' understanding of patients' needs and concerns. They learn to be empathetic and prioritize patient-centered care, enabling them to develop solutions that consider patients' needs and desires.
- Ethical awareness: Students are confronted with ethical issues related to digital health. They learn to analyze and evaluate ethical aspects and incorporate them into their decision-making process. This cultivates an awareness of the ethical challenges in the field of digital health and strengthens their ability to make responsible decisions.

Through the practical application of these social competencies in various exercises and group work, students are prepared to excel in the digital health sector and collaborate effectively with other professionals.

Autonomy

- Independent Learning: Students are encouraged to independently research knowledge, study additional literature, and engage with current developments in the digital health landscape. This cultivates their ability for self-directed learning and continuous professional development.
- Problem-solving Skills: By working on case studies and real-world problems in the field of digital health, students are prompted to think critically and develop solution-oriented approaches. They learn to analyze complex challenges, weigh different options, and make well-informed decisions.
- Time Management: The lecture requires students to plan and organize their time effectively in order to successfully complete various tasks, such as preparing presentations, engaging in group work, and working on case studies. They learn to set priorities, meet deadlines, and work efficiently.
- Critical Reflection: Students are encouraged to engage in critical thinking about the content and concepts of digital health. They learn to consider different perspectives, question their own assumptions, and construct their opinions based on well-founded arguments. This fosters a critical mindset and the ability for self-reflective practice.
- Self-Responsibility: Students are encouraged to take responsibility for their own learning and personal development. They learn to set their own goals, monitor their progress, and take initiative to continuously improve their knowledge and skills in the field of digital health.

Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** Yes · **Bonus** 20 % · **Form** Excercises · **Description** Erfolgreiche Teilnahme PBL-Übung
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory International Management and Engineering: Specialisation II. Medical Engineering: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory

Course L3099: Digital Health	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Moritz Göldner
Language	EN
Cycle	WiSe
Content	This course provides an in-depth exploration of the rapidly evolving field of digital health. It covers the current trends, state of the industry, and the perspectives of both patients and physicians, with particular emphasis on digital health applications (DiGA and DiPA) in Germany and Europe. Students will gain insights into the importance of interoperability, data management, and research data, while also exploring into the role of big data and AI in state-of-the-art healthcare. The course integrates theory with real-world application, case studies and a guest lecture, offering a comprehensive understanding of the digital transformation in the healthcare sector.
Literature	• Stern, A. D., Matthies, H., Hagen, J., Brönneke, J. B., & Debatin, J. F. (2020). Want to see the future of digital health tools? Look to Germany. Harvard Business Review, 2. https://hbr.org/2020/12/want-to-see-the-future-of-digital-health-tools-look-to-germany

Course L3100: Digital Health Seminar	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Moritz Göldner
Language	EN
Cycle	WiSe
Content	This course provides an in-depth exploration of the rapidly evolving field of digital health. It covers the current trends, state of the industry, and the perspectives of both patients and physicians, with particular emphasis on digital health applications (DiGA and DiPA) in Germany and Europe. Students will gain insights into the importance of interoperability, data management, and research data, while also exploring into the role of big data and AI in state-of-the-art healthcare. The course integrates theory with real-world application, case studies and a guest lecture, offering a comprehensive understanding of the digital transformation in the healthcare sector.
Literature	• Stern, A. D., Matthies, H., Hagen, J., Brönneke, J. B., & Debatin, J. F. (2020). Want to see the future of digital health tools? Look to Germany. Harvard Business Review, 2. https://hbr.org/2020/12/want-to-see-the-future-of-digital-health-tools-look-to-germany

Module M1739: Operational Aspekts in Aviation

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Flight Guidance I (Introduction) (L0848)	Lecture	2	2
Flight Guidance I (Introduction) (L0854)	Recitation Section (large)	1	1
Aviation and Environment (L2376)	Lecture	3	3

Module Responsible	Prof. Volker Gollnick
Admission Requirements	None
Recommended Previous Knowledge	Air Transportation Systems
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Analysis and description of the interaction between people and aircraft in operation
Skills	Understanding and application of design and calculation methods Understanding of interdisciplinary and integrative interdependencies Evaluation of operational issues in aviation and development of operational solution options
Personal Competence	
Social Competence	Working in teams for focused solutions communication, assertiveness, technical persuasion
Autonomy	Organisation of worksflows and strategies for solutions structured task analysis and definition of solutions
Workload in Hours	Depends on choice of courses
Credit points	6
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory

Course L0848: Flight Guidance I (Introduction)

Typ	Lecture
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Prof. Volker Gollnick
Language	DE
Cycle	WiSe
Content	Introduction and motivation Flight guidance principles (airspace structures, organization of air navigation services, etc.) Cockpit systems and Avionics (cockpit design, cockpit equipment, displays, computers and bus systems) Principles of flight measurement techniques (Measurement of position (geometric methods, distance measurement, direction measurement) Determination of the aircraft attitude (magnetic field- and inertial sensors) Measurement of speed Principles of Navigation Radio navigation Satellite navigation Airspace surveillance (radar systems) Commuication systems Integrated Navigation and Guidance Systems
Literature	Rudolf Brockhaus, Robert Luckner, Wolfgang Alles: "Flugregelung", Springer Berlin Heidelberg New York, 2011 Holger Flühr: "Avionik und Flugsicherungssysteme", Springer Berlin Heidelberg New York, 2013 Volker Gollnick, Dieter Schmitt "Air Transport Systems", Springer Berlin Heidelberg New York, 2016 R.P.G. Collinson „Introduction to Avionics”, Springer Berlin Heidelberg New York 2003

Course L0854: Flight Guidance I (Introduction)	
Typ	Recitation Section (large)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Examination Form	Klausur
Examination duration and scale	60 min
Lecturer	Prof. Volker Gollnick
Language	DE
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L2376: Aviation and Environment	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Examination Form	Klausur
Examination duration and scale	90 min
Lecturer	Dr. Florian Linke
Language	DE
Cycle	SoSe

Content	The lecture provides the necessary basics and methods for understanding the interactions between air traffic and the environment, both in terms of the effects of weather / climate on flying and with regard to the effects of air traffic on pollutant emissions, noise and climate.
	
	The following topics are covered:
	
	• Atmospheric physics / chemistry
	    ◦ Structure and statics
	    ◦ Dynamics (water cycle, formation of weather events, high and low pressure areas, wind, gusts and turbulence)
	    ◦ Cloud physics (thermodynamics, contrails)
	    ◦ Radiation physics (energy balance, greenhouse effect)
	    ◦ Photochemistry (ozone chemistry)
	• Impact of weather on flying
	    ◦ Atmospheric influences on flight performance
	    ◦ Flight planning
	    ◦ Disturbances due to weather, e.g. thunderstorms, winter weather (icing), clear air turbulence, visibility
	    ◦ Effects of climate change and adaptation
	• Effects of air traffic on the environment and climate
	    ◦ Aviation pollutant emissions
	    ◦ Effect of emissions on concentrations in the atmosphere
	    ◦ Climate metrics / models and background scenarios
	    ◦ Emissions inventories
	• Mitigation measures
	    ◦ Technological measures, e.g. climate-optimized aircraft design
	    ◦ Alternative fuels
	    ◦ Operational measures, e.g. climate-optimized flight planning
	    ◦ Environmental policy measures, e.g. EU-ETS, CORSIA
	    ◦ Potentials and comparison, concept of eco-efficiency
	• Local environmental impacts
	    ◦ Local air quality (particulate matter, other emissions near the ground)
	    ◦ Noise (noise sources, noise metrics, noise impact, measurement, certification, psychoacoustics, noise mitigation)
	    ◦ Health effects
	• Aspects of sustainability
	    ◦ Other aspects, including life cycle emissions, disposal/recycling
	    ◦ Relation to global goals, e.g. United Nations goals for sustainable development, Paris climate agreement

Literature	• Ruijgrok, G.: Elements of Aircraft Pollution, Delft University Press, 2005 • Friedrich, R., Reis, S.: Emissions of Air Pollutants, Springer 2004 • Janic, M.: The Sustainability of Air Transportation, Ashgate, 2007 • Schumann, U. (ed.): Atmospheric Physics: Background - Methods - Trends, Springer, Berlin, Heidelberg, 2012 • Spiridonov, V., Curic, M.: Fundamentals of Meteorology, Springer, 2021 • Kaltschmitt, M., Neuling, U.: Biokerosene - Status and Prospects, Springer, 2018 • Roedel, W., Wagner, T.: Physik unserer Umwelt: Die Atmosphäre, Springer, 2017 • W. Bräunling: Flugzeugtriebwerke. Springer-Verlag Berlin, Deutschland, 2009 • G. Brüning, X. Hafer, G. Sachs: Flugleistungen, Springer, 1993

Module M0623: Intelligent Systems in Medicine

Courses

Title	Typ	Hrs/wk	CP
Intelligent Systems in Medicine (L0331)	Lecture	2	3
Intelligent Systems in Medicine (L0334)	Project Seminar	2	2
Intelligent Systems in Medicine (L0333)	Recitation Section (small)	1	1

Module Responsible	Prof. Alexander Schlaefer
Admission Requirements	None
Recommended Previous Knowledge	• principles of math (algebra, analysis/calculus) • principles of stochastics • principles of programming, Java/C++ and R/Matlab • advanced programming skills
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students are able to analyze and solve clinical treatment planning and decision support problems using methods for search, optimization, and planning. They are able to explain methods for classification and their respective advantages and disadvantages in clinical contexts. The students can compare different methods for representing medical knowledge. They can evaluate methods in the context of clinical data and explain challenges due to the clinical nature of the data and its acquisition and due to privacy and safety requirements.
Skills	The students can give reasons for selecting and adapting methods for classification, regression, and prediction. They can assess the methods based on actual patient data and evaluate the implemented methods.
Personal Competence	
Social Competence	The students are able to grasp practical tasks in groups, develop solution strategies independently, define work processes and work on them collaboratively. The students can critically reflect on the results of other groups, make constructive suggestions for improvement and also incorporate them into their own work.
Autonomy	The students can assess their level of knowledge and document their work results. They can critically evaluate the results achieved and present them in an appropriate argumentative manner to the other groups.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** / **Bonus** / **Form** / **Description** Yes / 10 % / Presentation Yes / 10 % / Written elaboration
Examination	Written exam
Examination duration and scale	90 minutes
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory

Course L0331: Intelligent Systems in Medicine	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	- methods for search, optimization, planning, classification, regression and prediction in a clinical context - representation of medical knowledge - understanding challenges due to clinical and patient related data and data acquisition The students will work in groups to apply the methods introduced during the lecture using problem based learning.
Literature	Russel & Norvig: Artificial Intelligence: a Modern Approach, 2012 Berner: Clinical Decision Support Systems: Theory and Practice, 2007 Greenes: Clinical Decision Support: The Road Ahead, 2007 Further literature will be given in the lecture

Course L0334: Intelligent Systems in Medicine	
Typ	Project Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Course L0333: Intelligent Systems in Medicine	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M1249: Medical Imaging

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Medical Imaging (L1694)	Lecture	2	3
Medical Imaging (L1695)	Recitation Section (small)	2	3

Module Responsible	Prof. Tobias Knopp
Admission Requirements	None
Recommended Previous Knowledge	Basic knowledge in linear algebra, numerics, and signal processing
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	After successful completion of the module, students are able to describe reconstruction methods for different tomographic imaging modalities such as computed tomography and magnetic resonance imaging. They know the necessary basics from the fields of signal processing and inverse problems and are familiar with both analytical and iterative image reconstruction methods. The students have a deepened knowledge of the imaging operators of computed tomography and magnetic resonance imaging.
Skills	The students are able to implement reconstruction methods and test them using tomographic measurement data. They can visualize the reconstructed images and evaluate the quality of their data and results. In addition, students can estimate the temporal complexity of imaging algorithms.
Personal Competence	
Social Competence	Students can work on complex problems both independently and in teams. They can exchange ideas with each other and use their individual strengths to solve the problem.
Autonomy	Students are able to independently investigate a complex problem and assess which competencies are required to solve it.
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	None
Examination	Written exam
Examination duration and scale	90 min
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Computer Science in Engineering: Specialisation I. Computer Science: Elective Compulsory Interdisciplinary Mathematics: Specialisation Computational Methods in Biomedical Imaging: Compulsory Microelectronics and Microsystems: Specialisation Communication and Signal Processing: Elective Compulsory Technomathematics: Specialisation II. Informatics: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory

Course L1694: Medical Imaging

Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	• Overview about different imaging methods • Signal processing • Inverse problems • Computed tomography • Magnetic resonance imaging • Compressed Sensing • Magnetic particle imaging
Literature	**Bildgebende Verfahren in der Medizin**; O. Dössel; Springer, Berlin, 2000 **Bildgebende Systeme für die medizinische Diagnostik**; H. Morneburg (Hrsg.); Publicis MCD, München, 1995 **Introduction to the Mathematics of Medical Imaging**; C. L.Epstein; Siam, Philadelphia, 2008 **Medical Image Processing, Reconstruction and Restoration**; J. Jan; Taylor and Francis, Boca Raton, 2006 **Principles of Magnetic Resonance Imaging**; Z.-P. Liang and P. C. Lauterbur; IEEE Press, New York, 1999

Course L1695: Medical Imaging	
Typ	Recitation Section (small)
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Tobias Knopp
Language	EN
Cycle	WiSe
Content	See interlocking course
Literature	See interlocking course

Module M0630: Robotics and Navigation in Medicine

Courses

Title	Typ	Hrs/wk	CP
Robotics and Navigation in Medicine (L0335)	Lecture	2	3
Robotics and Navigation in Medicine (L0338)	Project Seminar	2	2
Robotics and Navigation in Medicine (L0336)	Recitation Section (small)	1	1

Module Responsible	Prof. Alexander Schlaefer
Admission Requirements	None
Recommended Previous Knowledge	• principles of math (algebra, analysis/calculus) • principles of programming, e.g., in Java or C++ • solid R or Matlab skills
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students can explain kinematics and tracking systems in clinical contexts and illustrate systems and their components in detail. Systems can be evaluated with respect to collision detection and safety and regulations. Students can assess typical systems regarding design and limitations.
Skills	The students are able to design and evaluate navigation systems and robotic systems for medical applications.
Personal Competence	
Social Competence	The students are able to grasp practical tasks in groups, develop solution strategies independently, define work processes and work on them collaboratively. The students are able to collaboratively organize their work processes and software solutions using virtual communication and software management tools. The students can critically reflect on the results of other groups, make constructive suggestions for improvement, and also incorporate them into their own work.
Autonomy	The students can assess their level of knowledge and independently control their learning processes on this basis as well as document their work results. They can critically evaluate the results achieved and present them in an appropriate argumentative manner to the other groups.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	**Compulsory** / **Bonus** / **Form** / **Description** Yes / 10 % / Written elaboration Yes / 10 % / Presentation
Examination	Written exam
Examination duration and scale	90 minutes
Assignment for the Following Curricula	Computer Science: Specialisation II: Intelligence Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Medical Technology: Elective Compulsory Electrical Engineering: Specialisation Medical Technology: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory International Management and Engineering: Specialisation II. Electrical Engineering: Elective Compulsory International Management and Engineering: Specialisation II. Process Engineering and Biotechnology: Elective Compulsory International Management and Engineering: Specialisation II. Medical Engineering: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Product Development: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Production: Elective Compulsory Mechanical Engineering - Product Development and Production: Specialisation Materials: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory Biomedical Engineering: Specialisation Artificial Organs and Regenerative Medicine: Elective Compulsory Biomedical Engineering: Specialisation Implants and Endoprostheses: Elective Compulsory Biomedical Engineering: Specialisation Medical Technology and Control Theory: Elective Compulsory Biomedical Engineering: Specialisation Management and Business Administration: Elective Compulsory Product Development, Materials and Production: Specialisation Product Development: Elective Compulsory Product Development, Materials and Production: Specialisation Production: Elective Compulsory Product Development, Materials and Production: Specialisation Materials: Elective Compulsory Theoretical Mechanical Engineering: Specialisation Bio- and Medical Technology: Elective Compulsory

Course L0335: Robotics and Navigation in Medicine	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	SoSe
Content	- kinematics - calibration - tracking systems - navigation and image guidance - motion compensation The seminar extends and complements the contents of the lecture with respect to recent research results.
Literature	Spong et al.: Robot Modeling and Control, 2005 Troccaz: Medical Robotics, 2012 Further literature will be given in the lecture.

Course L0338: Robotics and Navigation in Medicine	
Typ	Project Seminar
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Course L0336: Robotics and Navigation in Medicine	
Typ	Recitation Section (small)
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Schlaefer
Language	EN
Cycle	SoSe
Content	See interlocking course
Literature	See interlocking course

Module M1879: Causal Data Science for Business Analytics

Courses

Title	Typ	Hrs/wk	CP
Business Analytics with Causal Data Science (L3096)	Project-/problem-based Learning	3	3
Causal Data Science (L3095)	Lecture	2	3

Module Responsible	Prof. Christoph Ihl
Admission Requirements	None
Recommended Previous Knowledge	- Linear Algebra - Basics of programming - School knowledge in economics
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	After completing the module, students will be able to: - understand the difference between “correlation” and “causation”. - understand the shortcomings of current correlation-based approaches. - discuss the conceptual ideas behind various causal data science tools and algorithms. - critical examination of (study) results and spurious correlations. - understanding of application of methods in business and practice.
Skills	- develop causal knowledge relevant for specific data-driven decisions. - carry out state-of the art causal data analyses. - isolating causal effects despite the existence of confounding factors. - programming in relevant programming languages. - selection of the appropriate method depending on the problem.
Personal Competence	
Social Competence	Students can work on the problems both individually and in groups during the exercise times and also ask questions and contribute to the solution of other people's problems outside the exercise times in a dedicated forum for the course (Mattermost). In addition, students learn to prepare and present their results during the course.
Autonomy	Students learn to transfer the knowledge and skills they have learned to other subject areas and to link them to new learning content. To obtain information and solve problems, especially those related to programming errors, they learn to use appropriate resources to help themselves.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Solutions to coding problem sets after each class session
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory

Course L3096: Business Analytics with Causal Data Science	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Christoph Ihl
Language	EN
Cycle	SoSe
Content	Most managerial decision problems require answers to questions such as “what happens to Y if we do X?”, or “was it X that caused Y to change?” In other words, practical business decision-making requires knowledge about cause-and-effect. While most data science and machine learning approaches are designed to efficiently detect patterns in high-dimensional data, they are not able to distinguish causal relationships from simple correlations. That means, commonly used approaches to business analytics often fall short to provide decision makers with important causal knowledge. Therefore, many leading companies currently try to develop specific causal data science capabilities. This module will provide an introduction into the topic of causal inference with the help of modern data science and machine learning approaches and with a focus on applications to practical business problems from various management areas. Based on an overarching framework for causal data science, the course will guide students to detect sources of confounding influence factors, understand the problem of selective measurement in data collection, and extrapolate causal knowledge across different business contexts. We also cover several tools for causal inference, such as A/B testing and experiments, difference-in-differences, instrumental variables, matching, regression discontinuity designs, etc. A variety of hands-on examples will be discussed that allow students to apply their newly obtained knowledge and carry out state-of-the-art causal analyses by themselves. Topics covered: 1. Introduction and Overview 2. Probability and Regression Review 3. Potential Outcomes Causal Model 4. Directed Acyclic Graphs 5. Experiments and A/B-Testing 6. Matching and Subclassification 7. Regression Discontinuity 8. Instrumental Variables 9. Panel Data 10. Difference-in-Differences 11. Synthetic Control 12. Heterogeneous Treatment Effects 13. Mediation Analysis
Literature	• Angrist, J. D., & Pischke, J. S. (2014). Mastering metrics: The path from cause to effect. Princeton university press. • Cunningham, Scott (2021). Causal Inference: The Mixtape, New Haven: Yale University Press. • Hernán Miguel A., and Robins James M. (2020). Causal Inference: What If. Boca Raton: Chapman & Hall/CRC. • Huntington-Klein, Nick. The effect (2021). An introduction to research design and causality. Chapman and Hall/CRC. • Imbens, G. W., & Rubin, D. B. (2015). Causal inference in statistics, social, and biomedical sciences. Cambridge University Press. • Mullainathan, Sendhil, and Jann Spiess. (2017). Machine Learning: An Applied Econometric Approach. Journal of Economic Perspectives, 31(2): 87-106. • Pearl, Judea, and Dana Mackenzie (2018). The Book of Why. Basic Books, New York, NY. • Pearl, Judea, Madelyn Glymour, and Nicholas P. Jewell (2016). Causal Inference in Statistics: A Primer. John Wiley & Sons, Inc., New York, NY.

Course L3095: Causal Data Science	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Christoph Ihl
Language	EN
Cycle	SoSe
Content	Most managerial decision problems require answers to questions such as “what happens to Y if we do X?”, or “was it X that caused Y to change?” In other words, practical business decision-making requires knowledge about cause-and-effect. While most data science and machine learning approaches are designed to efficiently detect patterns in high-dimensional data, they are not able to distinguish causal relationships from simple correlations. That means, commonly used approaches to business analytics often fall short to provide decision makers with important causal knowledge. Therefore, many leading companies currently try to develop specific causal data science capabilities. This module will provide an introduction into the topic of causal inference with the help of modern data science and machine learning approaches and with a focus on applications to practical business problems from various management areas. Based on an overarching framework for causal data science, the course will guide students to detect sources of confounding influence factors, understand the problem of selective measurement in data collection, and extrapolate causal knowledge across different business contexts. We also cover several tools for causal inference, such as A/B testing and experiments, difference-in-differences, instrumental variables, matching, regression discontinuity designs, etc. A variety of hands-on examples will be discussed that allow students to apply their newly obtained knowledge and carry out state-of-the-art causal analyses by themselves. Topics covered: 1. Introduction and Overview 2. Probability and Regression Review 3. Potential Outcomes Causal Model 4. Directed Acyclic Graphs 5. Experiments and A/B-Testing 6. Matching and Subclassification 7. Regression Discontinuity 8. Instrumental Variables 9. Panel Data 10. Difference-in-Differences 11. Synthetic Control 12. Heterogeneous Treatment Effects 13. Mediation Analysis
Literature	• Angrist, J. D., & Pischke, J. S. (2014). Mastering metrics: The path from cause to effect. Princeton university press. • Cunningham, Scott (2021). Causal Inference: The Mixtape, New Haven: Yale University Press. • Hernán Miguel A., and Robins James M. (2020). Causal Inference: What If. Boca Raton: Chapman & Hall/CRC. • Huntington-Klein, Nick. The effect (2021). An introduction to research design and causality. Chapman and Hall/CRC. • Imbens, G. W., & Rubin, D. B. (2015). Causal inference in statistics, social, and biomedical sciences. Cambridge University Press. • Mullainathan, Sendhil, and Jann Spiess. (2017). Machine Learning: An Applied Econometric Approach. Journal of Economic Perspectives, 31(2): 87-106. • Pearl, Judea, and Dana Mackenzie (2018). The Book of Why. Basic Books, New York, NY. • Pearl, Judea, Madelyn Glymour, and Nicholas P. Jewell (2016). Causal Inference in Statistics: A Primer. John Wiley & Sons, Inc., New York, NY.

Module M1785: Machine Learning in Electrical Engineering and Information Technology

Courses

Title	Typ	Hrs/wk	CP
General Introduction Machine Learning (L3004)	Lecture	1	2
Machine Learning Applications in Electric Power Systems (L3008)	Lecture	1	1
Machine Learning in Electromagnetic Compatibility (EMC) Engineering (L3006)	Lecture	1	1
Machine Learning in High-Frequency Technology and Radar (L3007)	Lecture	1	1
Machine Learning in Wireless Communications (L3005)	Lecture	1	1

Module Responsible	Prof. Gerhard Bauch
Admission Requirements	None
Recommended Previous Knowledge	The module is designed for a diverse audience, i.e. students with different background. It shall be suitable for both students with deeper knowledge in machine learning methods but less knowledge in electrical engineering, e.g. math or computer science students, and students with deeper knowledge in electrical engineering but less knowledge in machine learning methods, e.g. electrical engineering students. Machine learning methods will be explained on a relatively high level indicating mainly principle ideas. The focus is on specific applications in electrical engineering and information technology. The chapters of the course will be understandable in different depth depending on the individual background of the student. The individual background of the students will be taken into consideration in the oral exam.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	The students know basic machine learning concepts and learning strategies. They are aware of specific opportunities, challenges and approaches of machine learning in various fields of electrical engineering. They know exemplary applications of machine learning in electrical engineering. The students are familiar with the contents of the module courses. They can explain and apply them to new problems.
Skills	The students are able to apply methods from machine learning to problems in electrical engineering. They are able to determine, dimension and implement suitable approaches such as types of deep learning networks and learning strategies for specific engineering problems. In particular, they are able to include domain knowledge in machine learning architectures and learning strategies. They are able to critically assess the learning results based on domain knowledge.
Personal Competence	
Social Competence	The students can jointly solve specific problems.
Autonomy	The students are able to acquire relevant information from appropriate literature sources. They can control their level of knowledge during the lecture period e.g. by solving tutorial problems or using software tools.
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	None
Examination	Oral exam
Examination duration and scale	30 min
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering and Information Technology: Specialisation Wireless and Sensor Technologies: Elective Compulsory Electrical Engineering: Specialisation Information and Communication Systems: Elective Compulsory Electrical Engineering: Specialisation Microwave Engineering, Optics, and Electromagnetic Compatibility: Elective Compulsory Electrical Engineering: Specialisation Control and Power Systems Engineering: Elective Compulsory Electrical Engineering: Specialisation Wireless and Sensor Technologies: Elective Compulsory Computer Science in Engineering: Specialisation II. Engineering Science: Elective Compulsory Information and Communication Systems: Specialisation Communication Systems, Focus Software: Elective Compulsory

Course L3004: General Introduction Machine Learning	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Dr. Maximilian Stark
Language	EN
Cycle	SoSe

Content	• From Rule-Based Systems to Machine Learning
	  ◦ Brief overview recent advances in ML in various domain
	  ◦ Outline and expected learning outcomes
	  ◦ Basics statistical inference and statistics
	  ◦ Basics of information theory
	• The Notions of Learning in Machine Learning
	  ◦ Unsupervised and supervised machine learning
	  ◦ Model-based and data-driven machine learning
	  ◦ Hybrid modelling
	  ◦ Online/offline/meta/transfer learning
	  ◦ General loss functions
	• Introduction to Deep Learning
	  ◦ Variants of neural networks
	  ◦ MLP
	  ◦ Conv. neural networks
	  ◦ Recurrent neural networks
	  ◦ Training neural networks
	  ◦ (Stochastic) Gradient Descent
	• Regression vs. Classification
	  ◦ Classification as supervised learning problem
	  ◦ Hands-On Session
	• Representation Learning and Generative Models
	  ◦ AutoEncoders
	  ◦ Directed Generative Models
	  ◦ Undirected Generative Models
	  ◦ Generative Adversarial Neural Networks
	• Probabilistic Graphical Models
	  ◦ Bayesian Networks
	  ◦ Variational inference (variational autoencoder)

Literature	Lecture notes as well as current literature announced in the lecture.

Course L3008: Machine Learning Applications in Electric Power Systems	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Becker, Dr. Davood Babazadeh
Language	EN
Cycle	SoSe
Content	This part of the course focuses on how to utilize ML methods to model and operate electric power systems. Electric power systems consist of generation units such as PV, loads or consumers and the grid that connects those actors and supports to transport energy. This part of the course helps to understand the data-driven modelling of generation units (e.g. PV & fuel cells), modelling of load behavior, and to formulate and solve a state estimation problem for distribution grids using neural networks. This part of the course includes lectures to introduce the basics that are followed by practical examples and coding.
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3006: Machine Learning in Electromagnetic Compatibility (EMC) Engineering	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Christian Schuster, Dr. Cheng Yang
Language	EN
Cycle	SoSe
Content	Electromagnetic Compatibility (EMC) Engineering deals with design, simulation, measurement, and certification of electronic and electric components and systems in such a way that their operation is safe, reliable, and efficient in any possible application. Safety is hereby understood as safe with respect to parasitic effects of electromagnetic fields on humans as well as on the operation of other components and systems nearby. Examples for components and systems range from the wiring in aircraft and ships to high-speed interconnects in server systems and wireless interfaces for brain implants. In this part of the course we will give an introduction to the physical basics of EMC engineering and then show how methods of Machine Learning (ML) can be applied to expand todays physcis-based approaches in EMC Engineering.
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3007: Machine Learning in High-Frequency Technology and Radar	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Prof. Alexander Kölpin
Language	EN
Cycle	SoSe
Content	Modern high-frequency systems benefit massively from machine learning methods. In applications where rule-based algorithms reach their limits, these data-driven approaches enable a significant increase in resolution and accuracy. This is exemplified by current research challenges, namely for the classification of targets in autonomous driving radar systems, radar-based gesture recognition for smart home applications and device control as well as in the field of medical technology for the contactless monitoring of human vital signs.
Literature	Lecture notes as well as current literature announced in the lecture.

Course L3005: Machine Learning in Wireless Communications	
Typ	Lecture
Hrs/wk	1
CP	1
Workload in Hours	Independent Study Time 16, Study Time in Lecture 14
Lecturer	Dr. Maximilian Stark
Language	EN
Cycle	SoSe

Content	• Supervised Learning Application - Channel Coding
	  ◦ Recap channel coding and block codes
	  ◦ Block codes as trainable neural networks
	  ◦ Tanner graph with trainable weights
	  ◦ Hands-on session
	• Supervised Learning Application - Modulation Detection
	  ◦ Recap wireless modulation schemes
	  ◦ Convolutional neuronal networks for blind detection of modulation schemes
	  ◦ Hands-on session
	• Autoencoder Application - Constellation Shaping I
	  ◦ Recap channel capacity and constellation shaping,
	  ◦ Capacity achieving machine learning systems
	  ◦ Information theoretical explanation of the autoencoder training
	  ◦ Hands-on session
	• Autoencoder Application - Constellation Shaping II
	  ◦ Training without a channel model
	  ◦ Mutual information neural estimator
	  ◦ Hands-on session
	• Generative Adversarial Network Application - Channel Modelling
	  ◦ Recap realistic channels with non-linear hardware impairments
	  ◦ Training a digital twin of a realistic channel with insufficient training data
	  ◦ Hands-on session
	• Recurrent Neural Network Application - Channel prediction
	  ◦ Recap time-varying channel models
	  ◦ Recurrent neural networks for temporal prediction
	  ◦ Hands-on session

Literature	Lecture notes as well as current literature announced in the lecture.

Module M1880: Deep Learning for Social Analytics

Courses

Title	Typ	Hrs/wk	CP
Deep Learning for Text and Graphs (L3097)	Lecture	2	3
Social Analytics with Deep Learning (L3098)	Project-/problem-based Learning	3	3

Module Responsible	Prof. Christoph Ihl
Admission Requirements	None
Recommended Previous Knowledge	• Basic knowledge of Python • Familiarity with probability theory, linear algebra and statistics
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• Understand how text and graphs can be transformed into data • Identify underlying relational structures of data that can be represented as graphs • Discuss the conceptual ideas behind various deep learning architectures • Decide about suitable deep learning architectures for a given task
Skills	• Proficiency in Python for deep learning applications • Apply basic natural language processing methods such as embedding and dependency parsing • Model complex data using graph representations • Set up deep learning architectures for different tasks • Make predictions employing deep learning models
Personal Competence	
Social Competence	• Collaboration on projects and assignments • Communication regarding computational, algorithmic and modeling challenges
Autonomy	• Maneuver in the field of deep learning including scientific literature and models • Solve computational, algorithmic, and modeling challenges related to deep learning models • Critical thinking skills • Self-sufficient problem-solving regarding coding issues
Workload in Hours	Independent Study Time 110, Study Time in Lecture 70
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Solutions to coding problem sets after each class session
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory Data Science: Specialisation IV. Special Focus Area: Elective Compulsory International Management and Engineering: Specialisation II. Information Technology: Elective Compulsory

Course L3097: Deep Learning for Text and Graphs	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Christoph Ihl
Language	EN
Cycle	WiSe
Content	Today, massive amounts of valuable data come in digital, yet often unstructured forms such as text or graphs. People communicate almost everything in language: e.g., social media, web search, product reviews, advertising, emails, customer service, language translation, chatbots, medical reports, etc. At the same time, they choose to interact with other people, products or websites. These networked interaction patterns can be represented as graphs of relationships between people and objects. Analyzing these new data sources and forms can help decision makers to significantly improve the effectiveness and efficiency of products, services and processes. This course introduces the fundamentals and current state of machine learning for natural language processing (NLP) and graphs in terms of content, users, and social relations. The course has a particular emphasis on key advancements in deep learning (or neural network) architectures, which in recent years have obtained very high performance across many different tasks, using single end-to-end models that do not require traditional, task-specific feature engineering. The course focuses on the computational, algorithmic, and modeling challenges specific to learning architecture for text and graphs. Students will gain a thorough introduction to modern deep learning algorithms. Through lectures and coding labs, students will learn the necessary skills to design, implement, and understand their own deep learning models. We will use Python and the deep learning framework PyTorch (Geometric). Topics Covered: 1. Intro: Text and Graphs as Data 2. Word Embeddings 3. Fundamentals of Deep Learning 4. Dependency Parsing 5. Recurrent Neural Networks for Text 6. Contextual Word Embeddings with Transformers 7. Analyzing Graphs 8. Graph Embeddings 9. Graph Embeddings for Complex Graphs 10. Graph Neural Networks (GNNs) 11. GNNs for Complex Graphs 12. GNNs for Text 13. Deep Generative Models for Text and Graphs
Literature	• Chollet, F., & Allaire, J. J. (2018). Deep Learning mit R und Keras: Das Praxis-Handbuch von den Entwicklern von Keras und RStudio. MITP-Verlags GmbH & Co. KG. • Hamilton, William L. (2020). Graph Representation Learning. Synthesis Lectures on Artificial Intelligence and Machine Learning, Vol. 14, No. 3 , Pages 1-159. • Hapke, H., Howard, C., & Lane, H. (2019). Natural Language Processing in Action: Understanding, analyzing, and generating text with Python. Simon and Schuster. • Hvitfeldt, E., & Silge, J. (2021). Supervised machine learning for text analysis in R. • Ma, Y., & Tang, J. (2021). Deep learning on graphs. Cambridge University Press. • Rao, D., & McMahan, B. (2019). Natural language processing with PyTorch: build intelligent language applications using deep learning. O'Reilly Media, Inc.

Course L3098: Social Analytics with Deep Learning	
Typ	Project-/problem-based Learning
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Christoph Ihl
Language	EN
Cycle	WiSe
Content	Today, massive amounts of valuable data come in digital, yet often unstructured forms such as text or graphs. People communicate almost everything in language: e.g., social media, web search, product reviews, advertising, emails, customer service, language translation, chatbots, medical reports, etc. At the same time, they choose to interact with other people, products or websites. These networked interaction patterns can be represented as graphs of relationships between people and objects. Analyzing these new data sources and forms can help decision makers to significantly improve the effectiveness and efficiency of products, services and processes. This course introduces the fundamentals and current state of machine learning for natural language processing (NLP) and graphs in terms of content, users, and social relations. The course has a particular emphasis on key advancements in deep learning (or neural network) architectures, which in recent years have obtained very high performance across many different tasks, using single end-to-end models that do not require traditional, task-specific feature engineering. The course focuses on the computational, algorithmic, and modeling challenges specific to learning architecture for text and graphs. Students will gain a thorough introduction to modern deep learning algorithms. Through lectures and coding labs, students will learn the necessary skills to design, implement, and understand their own deep learning models. We will use Python and the deep learning framework PyTorch (Geometric). Topics Covered: 1. Intro: Text and Graphs as Data 2. Word Embeddings 3. Fundamentals of Deep Learning 4. Dependency Parsing 5. Recurrent Neural Networks for Text 6. Contextual Word Embeddings with Transformers 7. Analyzing Graphs 8. Graph Embeddings 9. Graph Embeddings for Complex Graphs 10. Graph Neural Networks (GNNs) 11. GNNs for Complex Graphs 12. GNNs for Text 13. Deep Generative Models for Text and Graphs
Literature	• Chollet, F., & Allaire, J. J. (2018). Deep Learning mit R und Keras: Das Praxis-Handbuch von den Entwicklern von Keras und RStudio. MITP-Verlags GmbH & Co. KG. • Hamilton, William L. (2020). Graph Representation Learning. Synthesis Lectures on Artificial Intelligence and Machine Learning, Vol. 14, No. 3 , Pages 1-159. • Hapke, H., Howard, C., & Lane, H. (2019). Natural Language Processing in Action: Understanding, analyzing, and generating text with Python. Simon and Schuster. • Hvitfeldt, E., & Silge, J. (2021). Supervised machine learning for text analysis in R. • Ma, Y., & Tang, J. (2021). Deep learning on graphs. Cambridge University Press. • Rao, D., & McMahan, B. (2019). Natural language processing with PyTorch: build intelligent language applications using deep learning. O'Reilly Media, Inc. • Silge, J., & Robinson, D. (2017). Text mining with R: A tidy approach. O'Reilly Media, Inc.

Module M1402: Machine Learning in Logistics

Courses

Title	Typ	Hrs/wk	CP
Digitalization in Traffic and Logistics (L2004)	Lecture	1	2
Basics of Machine Learning (L2003)	Lecture	1	2
Machine Learning in Logistics (L2005)	Recitation Section (small)	2	2

Module Responsible	Prof. Carlos Jahn
Admission Requirements	None
Recommended Previous Knowledge	None
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	Students understand specific methods of machine learning. They are able to select appropriate procedures for given data. They can explain the principals of different learning methods. In addition, they can explain the major conceptual differences of learning methods.
Skills	Students can inspect, describe, and apply selected machine learning techniques to provided data sets. Additionally they can prepare raw data for machine learning algorithms. They are able to evaluate the usability in concrete company-relevant contexts and they know how to derive the requirements and potentials of an effective application, e.g. in relation to controlling or forecasting for the operational planning of companies or other organizations.
Personal Competence	
Social Competence	Students are capable of: • Discussing and organizing extensive research tasks in small groups • Jointly describing, differentiating between and evaluating problems
Autonomy	Students are able: • To research and select specialized literature • Read existing code, interpret it and modify it for new tasks
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory** / **Bonus** / **Form** / **Description** No / 15 % / Presentation
Examination	Written exam
Examination duration and scale	90 minutes
Assignment for the Following Curricula	Data Science: Specialisation III. Applications: Elective Compulsory International Management and Engineering: Specialisation II. Logistics: Elective Compulsory Logistics, Infrastructure and Mobility: Specialisation Infrastructure and Mobility: Elective Compulsory Logistics, Infrastructure and Mobility: Specialisation Production and Logistics: Elective Compulsory Mechatronics: Core Qualification: Elective Compulsory

Course L2004: Digitalization in Traffic and Logistics	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Carlos Jahn
Language	EN
Cycle	WiSe
Content	When dealing with large amounts of data (big data), it is no longer possible for humans to spot all relevant data by simply looking at the raw data. In the context of logistics, the handling of temporal data and movement data plays a particularly important role. In this course the visualization, the calculation of statistics, and the application of machine learning algorithms are covered. Students are given various tools for later practical application. The course utilizes the machine learning methods learned in "Basics of Machine Learning". These are used and evaluated in the context of practical application in the field of traffic and logistics. In addition, various pre-processing steps for raw data are presented and it is discussed, under which conditions these measurements are applicable. The lecture contents are: • The project structure for Machine Learning in science and industry • Use cases for Machine Learning in logistics • ML-based Computer Vision in road traffic and logistics • Temporal and movement data in traffic • Automated anomaly detection
Literature	• Géron, Aurélien (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and Tensorflow: Concepts, Tools, and Techniques to Build Intelligent Systems. O'Reilly. • VanderPlas, Jake (2023). Python Data Science Handbook: Essential Tools for Working with Data. O'Reilly. • Chapman, Peter and Clinton, Janet and Kerber, Randy and Khabaza, Tom and Reinartz, Thomas and Russel H. Shearer, C and Wirth, Robert (2000). CRISP-DM 1.0 : Step-by-step data mining guide. • Ayyadevara, V Kishore; Reddy, Yeshwanth (2024). Modern Computer Vision with PyTorch: A practical roadmap from deep learning fundamentals to advanced applications and Generative AI. pact. • Aggarwal, Charu C. (2017). Outlier Analysis. Springer International Publishing Switzerland.

Course L2003: Basics of Machine Learning	
Typ	Lecture
Hrs/wk	1
CP	2
Workload in Hours	Independent Study Time 46, Study Time in Lecture 14
Lecturer	Prof. Sibylle Schupp
Language	EN
Cycle	WiSe

Content	Students are able to understand specific procedures of machine learning and to use on real life examples. Students are able to use appropriate procedures for given data.
	
	Students are able to explain the differences between instance and model based learning approaches and are able to use specific approaches in machine learning on the base of static and incremental growing data.
	
	By the use of uncertainty the students can explain how axioms, parameter or structures can be learned. Additional the students learn to develop different cluster techniques.
	
	Planned content:
	• Supervised Learning:
	  ◦ Regressions
	  ◦ Decision trees
	  ◦ Bayesian networks
	  ◦ K-next neighbors
	  ◦ Logistical regressions
	  ◦ Neuronal Networks
	  ◦ Support Vector Machines
	  ◦ Ensemble Learning
	• Unsupervised Learning:
	  ◦ Hierarchical Clustering, K-Mean

Literature	John D. Kelleher, Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies (MIT Press) Tom M. Mitchell, Machine Learning Kevin P. Murphy, Machine Learning: A Probabilistic Perspective

Course L2005: Machine Learning in Logistics	
Typ	Recitation Section (small)
Hrs/wk	2
CP	2
Workload in Hours	Independent Study Time 32, Study Time in Lecture 28
Lecturer	Prof. Carlos Jahn
Language	EN
Cycle	WiSe
Content	In the exercise, the skills which the students acquired in the lectures will be applied to real life examples.
Literature	• Aggarwal, Charu C. (2017). Outlier Analysis. Springer International Publishing Switzerland. • Chapman, Peter and Clinton, Janet and Kerber, Randy and Khabaza, Tom and Reinartz, Thomas and Russel H. Shearer, C and Wirth, Robert (2000). DM 1.0 : Step-by-step data mining guide. • Géron, Aurélien (2018). Praxiseinstieg Machine Learning mit Scikit-Learn und TensorFlow: Konzepte, Tools und Techniken für intelligente Systeme. O'Reilly. • Haneke, Uwe and Trahasch, Stephan and Zimmer, Michael and Felden, Carsten (2019). Data Science - Grundlagen, Architekturen und Anwendungen. dpunk • Kelleher, John D. (2015) Fundamentals of Machine Learning for Predictive Data Analytics: Algorithms, Worked Examples, and Case Studies. MIT Press. • Mitchell, Tom M. (2005) Machine Learning. McGraw-Hill. • Murphy, Kevin P. (2012) Machine Learning: A Probabilistic Perspective. MIT Press. • VanderPlas, Jake (2017). Data Science mit Python : das Handbuch für den Einsatz von IPython, Jupyter, NumPy, Pandas, Matplotlib, Scikit-Learn. MIT Press.

Module M1807: Machine Learning for Physical Systems

Courses

Title	Typ	Hrs/wk	CP
Machine Learning for Physical Systems (L2987)	Lecture	2	3
Machine Learning for Physical Systems (L2988)	Project-/problem-based Learning	2	3

Module Responsible	Prof. Roland Can Aydin
Admission Requirements	None
Recommended Previous Knowledge	Prior knowledge in machine learning or Python programming is highly recommended, in particular a certain level of experience with standard ML libraries in Python (preferably PyTorch). No prior knowledge of specialized ML architectures, such as PINNs or large language models, is necessary.
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	In this module, students will delve into the advanced integration of machine learning techniques with physical systems. The course covers sophisticated topics, demonstrating how cutting-edge machine learning methodologies can be applied not only in general domains but are specifically tailored for complex physical systems. Core areas of study include: - Advanced Data Management**:** How can domain knowledge relating to physical problems be integrated into the pre- and postprocessing of data? - Transformer Architectures: Understanding the design and application of transformer models and large language models, focusing on their suitability to physical systems (e.g., Foundation models) - Physics-Informed Neural Networks: Architectures for embedding physical laws into a neural network's loss function - Constitutive Artificial Neural Networks: Architectures for embedding physical laws within a neural network's topology - Feature selection and dimensionality reduction - ML for Molecular Dynamics and Simulation - Synthetic Data Generation, particularly its usage to augment physical experiments (which are often a bottleneck in data generation) - Optimal Experimental Design: Techniques for efficiently gathering data through intelligently designed experiments. - Process-Structure-Properties Pipelines: Exploring specialised microstructural descriptors such as Gram-matrices to connect structure to either process parameters or mechanical properties Complementing the lectures, the associated exercise sessions will use various Python libraries such as Sklearn and Pytorch, typically within Jupyter notebooks. These practical sessions are designed to reinforce the concepts discussed in the lectures, with a reciprocal relationship between the theoretical and practical aspects of the course. This course is designed for those looking to understand and apply machine learning in the realm of physical systems, bridging the gap between abstract algorithms and real-world physical phenomena. The course is offered fully in English.
Skills	The students will be able to competently evaluate suitable machine learning methods for a given problem involving physical systems, understanding the advantages and disadvantages of each approach. They will be able to do so both for standard machine learning tools and methods as well as for specialised models.
Personal Competence	
Social Competence	The students will be able to reason for and against solutions for complex problems involving physical systems and to present their conclusions on how to incorporate their domain knowledge to facilitate the choice, design, training, and validation of an appropriate machine learning algorithm.
Autonomy	The module places a particular emphasis on enabling students to achieve the competence level both in group work (homework assignments) as well as individually (during the exercises).
Workload in Hours	Independent Study Time 124, Study Time in Lecture 56
Credit points	6
Course achievement	**Compulsory**: No; **Bonus**: 20 %; **Form**: Excercises; **Description**: Im Rahmen der Übung und über Stud.IP werden wöchentliche Übungsaufgaben bereitgestellt, durch deren korrekte Abgabe bis zu 20% als Bonus zur Abschlussprüfung erbracht werden können.
Examination	Written exam
Examination duration and scale	75 min
Assignment for the Following Curricula	General Engineering Science (German program, 7 semester): Specialisation Data Science: Elective Compulsory General Engineering Science (German program, 7 semester): Specialisation Advanced Materials: Compulsory General Engineering Science (German program, 7 semester): Specialisation Mechanical Engineering, Focus Mechatronics: Elective Compulsory General Engineering Science (German program, 7 semester): Specialisation Mechanical Engineering, Focus Theoretical Mechanical Engineering: Elective Compulsory Computational Methods and Machine Learning in Engineering: Core Qualification: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Energy Systems: Core Qualification: Elective Compulsory Engineering Science: Specialisation Advanced Materials: Compulsory Engineering Science: Specialisation Data Science: Elective Compulsory Aeronautics: Core Qualification: Elective Compulsory

	Materials Science and Engineering: Specialisation Modeling: Elective Compulsory Mechatronics: Specialisation Dynamic Systems and AI: Elective Compulsory ~~Mechatronics: Specialisation Robot- and Machine Systems: Elective Compulsory~~

Course L2987: Machine Learning for Physical Systems	
Typ	Lecture
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Roland Can Aydin
Language	EN
Cycle	WiSe
Content	In this lecture, students will delve into the advanced integration of machine learning techniques with physical systems. The lecture covers sophisticated topics, demonstrating how cutting-edge machine learning methodologies can be applied not only in general domains but are specifically tailored for complex physical systems. Core areas of study include: - Advanced Data Management**:** How can domain knowledge relating to physical problems be integrated into the pre- and postprocessing of data? - Transformer Architectures: Understanding the design and application of transformer models and large language models, focusing on their suitability to physical systems (e.g., Foundation models) - Physics-Informed Neural Networks: Architectures for embedding physical laws into a neural network's loss function - Constitutive Artificial Neural Networks: Architectures for embedding physical laws within a neural network's topology - Feature selection and dimensionality reduction - ML for Molecular Dynamics and Simulation - Synthetic Data Generation, particularly its usage to augment physical experiments (which are often a bottleneck in data generation) - Optimal Experimental Design: Techniques for efficiently gathering data through intelligently designed experiments. - Process-Structure-Properties Pipelines: Exploring specialised microstructural descriptors such as Gram-matrices to connect structure to either process parameters or mechanical properties This lecture is designed for those looking to understand and apply machine learning in the realm of physical systems, bridging the gap between abstract algorithms and real-world physical phenomena. The lecture is offered fully in English.
Literature	Relevante Literatur basiert vor allem auf wissenschaftlichen Veröffentlichungen (statt Lehrbüchern), die jeweiligen Referenzen werden in der Vorlesung bzw. Übung genannt.

Course L2988: Machine Learning for Physical Systems	
Typ	Project-/problem-based Learning
Hrs/wk	2
CP	3
Workload in Hours	Independent Study Time 62, Study Time in Lecture 28
Lecturer	Prof. Roland Can Aydin
Language	EN
Cycle	WiSe
Content	The exercise (PBL) demonstrates the methods introduced in the lecture on different example applications, focusing on gaining practical hands-on proficiency. By submitting correctly solved homework assignments, points can be earned for the module examination. Topics correspond to those presented at that time in the module's lecture.
Literature	Keine über die in der Vorlesung genannten Referenzen herausgehende Literatur ist notwendig.

Module M2076: Introduction to Climate Informed Engineering

Courses			
Title	**Typ**	**Hrs/wk**	**CP**
Methods in Climate Informed Engineering (L3347)	Lecture	3	3
Topics in Climate Informed Engineering (L3348)	Lecture	3	3

Module Responsible	Prof. Nima Shokri
Admission Requirements	None
Recommended Previous Knowledge	Students are expected to have a foundational understanding of environmental science, basic engineering principles, and an interest in sustainability. Recommended knowledge includes climate science, data analysis, and familiarity with engineering design processes. Analytical and critical thinking and creative problem-solving skills are also beneficial
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	This module explores next-generation climate models and high-resolution data, emphasizing their impact on environmental and engineering products and processes. It covers how various engineering disciplines can benefit from climate information. Research-based learning activities, expert talks, and presentations will expose students to state-of-the-art modeling, measurement, and analysis in climate-informed engineering.
Skills	Climate data analysis, engineering adaptation strategies, problem-solving, research-based learning, and interdisciplinary collaboration.
Personal Competence	
Social Competence	Collaboration, interdisciplinary teamwork, communication skills, problem-solving, ethical responsibility, and decision-making in climate-resilient engineering.
Autonomy	Time management, self-directed learning, critical thinking, accountability, initiative, and the ability to conduct independent research and make informed decisions in climate-informed engineering.
Workload in Hours	Independent Study Time 96, Study Time in Lecture 84
Credit points	6
Course achievement	None
Examination	Subject theoretical and practical work
Examination duration and scale	Report and Presentation
Assignment for the Following Curricula	Civil Engineering: Specialisation Coastal Engineering: Elective Compulsory Civil Engineering: Specialisation Geotechnical Engineering: Elective Compulsory Civil Engineering: Specialisation Structural Engineering: Elective Compulsory Civil Engineering: Specialisation Water and Traffic: Elective Compulsory Civil Engineering: Specialisation Computational Engineering: Elective Compulsory Data Science: Specialisation III. Applications: Elective Compulsory Environmental Engineering: Core Qualification: Elective Compulsory Process Engineering: Specialisation Process Engineering: Elective Compulsory Water and Environmental Engineering: Specialisation Cities: Elective Compulsory Water and Environmental Engineering: Specialisation Environment: Elective Compulsory Water and Environmental Engineering: Specialisation Water: Elective Compulsory

Course L3347: Methods in Climate Informed Engineering

Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Nima Shokri, Prof. Cathy Hohenegger, Prof. Irina Smirnova
Language	EN
Cycle	WiSe
Content	Students will learn techniques for incorporating climate data and environmental factors into engineering design. It covers climate modelling and the use of sensors and devices to measure climate-related parameters and engineering processes. Students will have the opportunity to conduct their own measurements, analyze the collected data, and write a report on their findings. This hands-on experience will be assessed and contribute to their final grade.
Literature	

Course L3348: Topics in Climate Informed Engineering	
Typ	Lecture
Hrs/wk	3
CP	3
Workload in Hours	Independent Study Time 48, Study Time in Lecture 42
Lecturer	Prof. Irina Smirnova, Prof. Cathy Hohenegger, Prof. Nima Shokri
Language	EN
Cycle	WiSe
Content	Exploring specific applications of climate data in various engineering disciplines. Invited speakers will present their research and discuss the relevance of climate-informed engineering to their work. Additionally, there will be a segment on effective communication, covering how to give impactful presentations and write research papers. Students will also give presentations on their own class projects related to climate-informed engineering, applying the concepts they’ve learned. This hands-on experience will be assessed and contribute to their final grade.
Literature	

Thesis

Module M-002: Master Thesis

Courses			
Title	**Typ**	**Hrs/wk**	**CP**

Module Responsible	Professoren der TUHH
Admission Requirements	• According to General Regulations §21 (1): At least 60 credit points have to be achieved in study programme. The examinations board decides on exceptions.
Recommended Previous Knowledge	
Educational Objectives	After taking part successfully, students have reached the following learning results
Professional Competence	
Knowledge	• The students can use specialized knowledge (facts, theories, and methods) of their subject competently on specialized issues. • The students can explain in depth the relevant approaches and terminologies in one or more areas of their subject, describing current developments and taking up a critical position on them. • The students can place a research task in their subject area in its context and describe and critically assess the state of research.
Skills	The students are able: • To select, apply and, if necessary, develop further methods that are suitable for solving the specialized problem in question. • To apply knowledge they have acquired and methods they have learnt in the course of their studies to complex and/or incompletely defined problems in a solution-oriented way. • To develop new scientific findings in their subject area and subject them to a critical assessment.
Personal Competence	
Social Competence	Students can • Both in writing and orally outline a scientific issue for an expert audience accurately, understandably and in a structured way. • Deal with issues competently in an expert discussion and answer them in a manner that is appropriate to the addressees while upholding their own assessments and viewpoints convincingly.
Autonomy	Students are able: • To structure a project of their own in work packages and to work them off accordingly. • To work their way in depth into a largely unknown subject and to access the information required for them to do so. • To apply the techniques of scientific work comprehensively in research of their own.
Workload in Hours	Independent Study Time 900, Study Time in Lecture 0
Credit points	30
Course achievement	None
Examination	Thesis
Examination duration and scale	According to General Regulations
Assignment for the Following Curricula	Civil Engineering: Thesis: Compulsory Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Chemical and Bioprocess Engineering: Thesis: Compulsory Computational Methods and Machine Learning in Engineering: Thesis: Compulsory Computer Science: Thesis: Compulsory Data Science: Thesis: Compulsory Electrical Engineering and Information Technology: Thesis: Compulsory Electrical Engineering: Thesis: Compulsory Energy Systems: Thesis: Compulsory Environmental Engineering: Thesis: Compulsory Aircraft Systems Engineering: Thesis: Compulsory Global Innovation Management: Thesis: Compulsory Computer Science in Engineering: Thesis: Compulsory Information and Communication Systems: Thesis: Compulsory Interdisciplinary Mathematics: Thesis: Compulsory International Production Management: Thesis: Compulsory International Management and Engineering: Thesis: Compulsory Joint European Master in Environmental Studies - Cities and Sustainability: Thesis: Compulsory Logistics, Infrastructure and Mobility: Thesis: Compulsory Aeronautics: Thesis: Compulsory Mechanical Engineering - Product Development and Production: Thesis: Compulsory Materials Science and Engineering: Thesis: Compulsory

	Materials Science: Thesis: Compulsory Mechanical Engineering and Management: Thesis: Compulsory Mechatronics: Thesis: Compulsory Biomedical Engineering: Thesis: Compulsory Microelectronics and Microsystems: Thesis: Compulsory Product Development, Materials and Production: Thesis: Compulsory Renewable Energies: Thesis: Compulsory Naval Architecture and Ocean Engineering: Thesis: Compulsory Naval Architecture and Ocean Engineering: Thesis: Compulsory Ship and Offshore Technology: Thesis: Compulsory Theoretical Mechanical Engineering: Thesis: Compulsory Process Engineering: Thesis: Compulsory Water and Environmental Engineering: Thesis: Compulsory Certification in Engineering & Advisory in Aviation: Thesis: Compulsory